# Supplementary material

Cryptic genomic variation in a panmictic squid: an ancient inversion supergene and genome-wide signatures of positive selection in *Illex illecebrosus*

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## Supplementary methods

**Read processing and variant calling.** Raw reads for all 633 individuals were downloaded from ENA project PRJEB101095 and processed with grenepipe (Czech and Exposito-Alonso, 2022). Twelve alignments hold almost no reads (93 to 2,009,803 autosomal reads in `samtools idxstats`, against a median of 57,182,056, listed in `data/qc/empty_bams_12.tsv`), eleven from division 4X and one from 3K. They were excluded from every count and estimate. Analyses that were run on all 633 received only missing genotypes or no reads from them, so those results rest on the remaining 621. Two further individuals, SQ23HBB023 (2+/2+, division 6C) and SQ23HBB124 (2a/2a, 6A), were excluded afterwards as likely contaminated, and SQ233K026 (2a/2a in the principal components, 3K) was excluded last as a possible mixed sample (`data/qc/excluded_15.tsv`, see Genotype quality and karyotype groups). The two-component Gaussian mixture on the Z:autosome coverage ratio had called both female, because its female component is about four times broader than the male one (standard deviation 0.146 against 0.034) and so takes in ratios beyond the upper end of the male cluster. Both carry Z heterozygosity in the male range (0.0096 and 0.0036, against a female 95th percentile of 0.0014), Z coverage above every male (1.46 and 1.61, against a male maximum of 1.34), and the two most extreme scores on the second autosomal principal component. Counts and karyotype frequencies are for the remaining 618 (222 2a/2a, 299 2a/2+ and 97 2+/2+). All principal-component analyses, the karyotype-by-division and capture-depth tests and the inversion analyses below were run on 619 individuals and include SQ233K026. Genome-wide analyses run before the exclusion, namely the spectra, demography, sweep scans, recombination maps, geographic $F_{ST}$ scans and the anomalous-diversity screen, include the two (2 of 621 individuals). Of the inversion analyses, only the pooled linkage-disequilibrium map (fig. S8) includes them. The arrangement statistics, fixed differences and the inversion model fits use the karyotype groups of the final 619 (223 2a/2a and 97 2+/2+ for the between-arrangement comparisons). Reads were trimmed with fastp (Chen et al., 2018) and mapped to the `Illex_F24` primary assembly with BWA-MEM2 (Vasimuddin et al., 2019), and duplicates were removed with Picard MarkDuplicates. Each individual was genotyped with GATK HaplotypeCaller in gVCF mode, and the gVCFs were jointly genotyped with all sites retained (Poplin et al., 2017). Variant records were annotated with snpEff and soft-filtered with bcftools (Danecek et al., 2021). A site was removed if its quality was below 20, its root-mean-square mapping quality was below 40, or its strand-bias score (FS) exceeded 60. It was also removed if it was missing in half or more of the individuals, if it lay in

a GC-corrected high-depth window identified with mosdepth, or if it lay outside the accessibility mask. Sites at intermediate allele frequency (0.1 to 0.9) with a strong excess of heterozygotes (ExcHet $< 0.001$) were flagged as likely paralogous and removed. Only sites passing every filter were used. The accessibility mask is the 1.43 Gb of the 3.47-Gb assembly that lies outside repeats annotated with Earl Grey. Mappability was computed with GenMap 1.3.0 (Pockrandt et al., 2020) for 150-mers with up to two mismatches, matching the 151-bp reads. Across the assembly 14.7% of positions are not uniquely mappable, but within the accessibility mask only 3.8% (53.7 Mb) are, because the repeat annotation already removes nearly all multi-copy sequence. Non-unique sequence is concentrated six-fold in the 22 anomalous-diversity blocks (23% of their accessible sequence), and outside them it is 2.7% of accessible sequence. For the sweep scan and the gene-set analyses, the mask was extended by the 22 anomalous-diversity blocks (129 Mb, Supplementary Table S5), in which collapsed duplications and sequence duplicated within the assembly make genotype-based statistics unreliable.

**Diversity and demography.** All site-frequency spectra and diversity statistics were computed with ANGSD 0.940 (Korneliussen et al., 2014) directly from the deduplicated BAM alignments of the 621 individuals with data, not from the called genotypes, so that the uncertainty of genotypes at about 2.5-fold depth (a median of 57.9 million autosomal reads of 150 bp per individual) is carried through as genotype likelihoods. Each chromosome was run separately with the options `-doSaf 1 -doCounts 1 -GL 1 -minQ 20 -minMapQ 20 -remove_bads 1 -only_proper_pairs 1 -minInd 1 -setMinDepthInd 1`, restricted with `-sites` to the accessibility mask and with the reference supplied as both `-ref` and `-anc`. `-GL 1` is the SAMtools genotype-likelihood model, and because the reference is not an ancestral sequence all spectra are folded. The folded spectrum was estimated with `realSFS -fold 1 -tole 1e-8 -maxIter 200`, per-site thetas with `realSFS saf2theta -fold 1`, and $\pi$, Watterson's $\theta$ and Tajima's $D$ with `thetaStat do_stat`, per chromosome and in 50-kb windows (10-kb step genome-wide, 50-kb step per division). The genome-wide folded spectrum pooled across chromosomes is the input to the demographic models. Per-division statistics used the same options on at most 25 individuals per division (all individuals in 3O, 4S and 4T), and the Z was run in the 329 confidently sexed males with data. Per-karyotype spectra and genotype-likelihood $F_{ST}$ between arrangements used the 2a/2a and 2+/2+ homozygotes as separate sample sets. Stairway Plot 2, moments (a two-epoch growth model fitted to the folded spectrum, with Godambe confidence intervals), and GONE2 were fitted as described in Supplementary Table S10, and the moments model was checked by forward simulation, which reproduced the observed $\pi$ and Tajima's $D$.

The coalescent class was tested by comparing the normalised folded genome-wide SFS, projected to 40 chromosomes, with expected shapes simulated in msprime (100 replicates of 2 Mb, branch-length spectra) under a constant-size Kingman coalescent, Kingman with the fitted growth history, and Beta$(2-\alpha, \alpha)$ coalescents without growth for $\alpha$ of 1.05 to 1.99 (`beta_vs_kingman.py` in `pipeline/steps/15_coalescent_model`). Fits were scored by $L_1$, the summed absolute difference between observed and expected proportions across the 20 frequency classes, because with $8.5 \times 10^7$ projected sites multinomial likelihood differences are dominated by sample size. $L_1$ is 0.036 for Kingman with growth (singleton fraction 0.481 against 0.483 observed), 0.108 for the best Beta coalescent ($\alpha = 1.35$, singleton fraction 0.516) and 0.537 for constant Kingman.

**Genotype-based statistics.** Statistics that need individual genotypes were computed with pg_gpu (Pope et al., 2026) from the filtered VCF. Windowed $\pi$, $d_{xy}$ and Hudson $F_{ST}$ between arrangements used biallelic SNPs in 100-kb windows, with the 2a/2a ($n = 223$) and 2+/2+ ($n = 97$)

homozygotes as the two populations and missing genotypes included pairwise. The between-division $F_{ST}$ matrix (Hudson), the linkage-disequilibrium triangles (genotype $r^2$ among common SNPs), and the principal-component analyses were computed the same way. The per-SNP geographic $F_{ST}$ scans (Supplementary Figs. S3, S4) used plink2 (Chang et al., 2015) to filter and count alleles at every biallelic SNP with a minor-allele frequency above 0.05 in the pooled 620 individuals from the ten divisions with more than one sample, the 618 retained and the two excluded afterwards (division 3N, with a single individual, was excluded). Per-division allele counts were converted to Hudson $F_{ST}$ for each of the 45 division pairs (Bhatia et al., 2013), at sites genotyped in at least 40 individuals in that pair and with at least two alleles in each division. The per-SNP value is the ratio of averages over informative pairs (the sum of the numerators over the sum of the denominators), without clipping, keeping sites informative in at least 20 of the 45 pairs. Of the 23,892,020 SNPs passing the frequency filter, 20,276,930 met this criterion, 20,117,984 of them on the 45 autosomes (133,957 in the chromosome-2 inversion) and 158,946 on the Z chromosome. The null was built by shuffling division labels among the individuals 20 times (division sizes kept) and recomputing $F_{ST}$ at the same SNPs in the same way. Thresholds are quantiles of the pooled null over the autosomes outside the chromosome-2 inversion ($3.996 \times 10^8$ values), and the expected count above each threshold is the number of SNPs times the null tail rate.

Between-division $F_{ST}$ values (main-text Fig. 6c) are slightly negative and scale with the inverse of the division sample sizes (Pearson $r = -0.98$ with $1/n_1 + 1/n_2$), so they reflect sample size rather than structure. Because Tajima's $D$ depends on sample size under population growth, the division medians in Fig. 6 ($-1.26$ to $-1.64$, at most 25 individuals per division) are not directly comparable to the genome-wide $D$ of $-2.07$ from all individuals, and projecting the genome-wide spectrum to 25 individuals gives $D = -1.50$.

**Genotype quality and karyotype groups.** At about 2.5-fold depth, heterozygotes are undercalled. On chromosome 20, over the biallelic PASS SNPs of the 619 individuals, the number of called heterozygotes is 0.36 of that expected under Hardy–Weinberg from the called allele frequencies (0.34 at minor-allele frequency $\geq$0.05, where 66% of genotypes are called), with $H_e = 2pq\,2n/(2n-1)$ per site from the $n$ called genotypes and $H_o/H_e = \sum H_o / \sum H_e$ over sites. Per individual, $H_o/H_e$ rises with read count from 0.14 in the lowest to 0.48 in the highest fifth (Spearman $\rho = 0.99$), and so does the call rate ($\rho = 0.99$, fig. S27). The same estimator gives $H_o/H_e = 0.32$ for each arrangement in the inversion and sets the heterozygote miscall rate $h = 1 - H_o/H_e$ of the genotype-error model in the simulations. The three karyotype groups do not differ in read count (medians 56.8, 59.8 and 53.7 million for 2a/2a, 2a/2+ and 2+/2+, Kruskal–Wallis $p = 0.13$), chromosome-20 call rate ($p = 0.09$) or $H_o/H_e$ ($p = 0.13$), and karyotype is independent of division ($\chi^2$ $p = 0.72$, the ten divisions with at least five individuals). With 97 2a/2a individuals matched to the 97 2+/2+ on read count (medians 53.5 and 53.7 million), $\pi_I/\pi_S$ is $0.708 \pm 0.027$ and $d_{xy}/\pi_S$ is $1.403 \pm 0.023$ over the same 189 windows, against $0.719 \pm 0.027$ and $1.414 \pm 0.024$ for all individuals, and from the ANGSD genotype likelihoods of the two groups $\pi_I/\pi_S$ is $0.704 \pm 0.026$. In the collinear flanks of chromosome 2 the groups have the same $\pi$ (ratio 1.00, and 0.98 when matched on depth). Karyotypes were also called directly from ANGSD genotype likelihoods at the 3,329 biallelic near-fixed sites between the breakpoints ($|\Delta\mathrm{AF}| \geq 0.9$), with genotype probabilities from the allele frequencies of each arrangement and a Hardy–Weinberg prior at $p = 0.602$. Every individual has a posterior above 0.99 (reads at a median of 2,815 sites), and the calls agree with the final karyotypes for 618 of the 619, including the two individuals whose full-depth and depth-equalised calls differ. The exception, SQ233K026, is 2a/2+ from the genotype likelihoods and 2a/2a from both principal-component fits. It lies at the edge of the 2a/2a cluster, carries inversion heterozygosity typical of

2a/2+ (0.083), and is one of the three female calls set to not confident for Z coverage between the clusters and male-like Z heterozygosity, so it may be a mixed sample, and it was excluded from the counts. The 40 individuals of each homokaryotype used for the joint spectra differ in sampling event (33 of the 2a/2a from division 3K against 20 of the 2+/2+), which contributes to the excess of private rare variants in 2+/2+ in the collinear control. Division-balanced draws of 40 + 40 reduce the ratio of the two private-rare cells from 1.42 to 1.18–1.22 on the same 10-Mb collinear block (chr2:85–95 Mb), so sampling composition accounts for about half of the asymmetry (`pipeline/steps/03_karyotype/m4_qc/`, `jsfs_balance.sh`).

**The inversion.** Karyotypes were called with a three-component Gaussian mixture on the first principal component of a PCA of chr2:60–80 Mb, fitted on the 619 individuals with the biallelic SNPs of minor-allele frequency above 0.05 among them (84,218 SNPs). To control for sequencing depth, every alignment was downsampled to about 1× in chr2:60–80 Mb (samtools, fixed seed), re-genotyped (77,209 SNPs), and karyotyped again in the same way. The two assignments agree for 617 of the 619 individuals (99.7%). The two disagreements, full-depth 2a/2a and 2+/2+ calls that are 2a/2+ at equal depth, have read depth near the median and inversion heterozygosity typical of 2a/2+ (0.062 and 0.070, against medians of 0.078 in 2a/2+ and 0.033 and 0.040 in the homozygotes). The final calls take the equal-depth assignment for both (`data/qc/karyotypes_619.tsv`), giving 223 2a/2a, 299 2a/2+ and 97 2+/2+ (222 2a/2a after the exclusion of SQ233K026). Breakpoints were placed at the step change in between-arrangement differentiation and confirmed by a strand flip in the AnchorWave alignment to *I. coindetii*. The arrangement carried by each haplotype of the reference individual was read from minimap2 2.29 (`-x asm5`) alignments of chromosome 2 of the two haplotype assemblies to each other and to the primary assembly, and the orientation of each haplotype relative to *I. coindetii* from alignments to the homologous *I. coindetii* chromosome (OZ346549.1, `-x asm20`). Alignment blocks of at least 1 Mb were ordered along each *Illex illecebrosus* haplotype, and polarity was read from which arrangement keeps the order and adjacency of *I. coindetii* sequence across the rearranged segment (Supplementary Table S3). The same comparison was made with the chromosome-level assembly of *Sthenoteuthis oualaniensis* (Li et al., 2022), a member of the same family. Its chromosome-2 homolog, HiC_scaffold_1, is the scaffold that receives 265 of the 319 *Illex illecebrosus* chromosome-2 genes transferred with LiftOn (Chao et al., 2025), and chromosome 2 of each haplotype assembly was aligned to it with minimap2 (`-x asm20`). Because the two genera differ by many small rearrangements along the chromosome, polarity was read at each breakpoint from which flanking segments are joined, using alignments of at least 5 kb with mapping quality of at least 5 together with the order of the transferred genes. Allele sharing with *I. coindetii* was also scored at the 15,093 sites in the inversion where allele frequency differs by at least 0.7 between 2a/2a and 2+/2+ homozygotes, with the *I. coindetii* base read from the AnchorWave alignment of *I. coindetii* to the primary assembly, which carries 2+ in the inversion. Where the *I. coindetii* base was callable (14,650 sites), the majority allele of 2a matches it at 26.0% of sites, so 2+ carries the *I. coindetii* allele at most diagnostic sites. The 2a matches are consistent with variation that was segregating in the ancestral population and was captured by the inversion when it formed, possibly in more than one capture event, which would give the soft start discussed in the main text. Chromosome 2 of both outgroups is rearranged in this region, so the arrangement of the common ancestor away from the breakpoints cannot be determined. It may have been an intermediate rearranged form, and the polarity used here is the reading on which *I. coindetii* and *S. oualaniensis* agree at both breakpoints. Restricted to the 4,582 sites where *I. coindetii* differs from the reference base, 2a carries the *I. coindetii* allele at 65.0% (2,980) and 2+ at 35.0% (1,602), but this subset is selected for sites at which the reference, a 2+ haplotype, carries a non-*I. coindetii*

allele, so it is biased toward 2a matches. Reads of ten *I. argentinus* individuals (about 0.3× each) mapped to the primary assembly were counted at the same diagnostic sites (`bcftools mpileup`, mapping and base quality of at least 20). Reference bias was measured at 150,000 SNPs in the collinear control (10–30 Mb), where the *I. argentinus* non-reference read fraction tracks the *Illex illecebrosus* non-reference allele frequency $p$ as $0.018 + 1.02p$, and the frequency of 2a in *I. argentinus* was estimated by maximum likelihood from the read counts under this calibration. All ten individuals give the same estimate (0.27–0.30, pooled 0.284), rather than the values near 0, 0.5 and 1 expected if the inversion segregated in *I. argentinus*. At sites covered by at least four reads, 87% carry reads of only one allele, as expected if each site is fixed, so *I. argentinus* carries the 2+ allele at most diagnostic sites and the 2a allele at the rest. Its 2a alleles fall mainly at sites where *I. coindetii* also carries the 2a allele (0.68 of reads, against 0.24 where *I. coindetii* carries the 2+ allele), so they are best read as ancestral states shared by both species, and the inversion polymorphism is not shared with *I. argentinus*. The age and frequency trajectory were fitted with msinv as described in the next paragraphs. Unannotated, non-repeat sequence in the inversion was searched with DIAMOND `blastx` against seven cephalopod proteomes, with transposable elements from Earl Grey (Baril et al., 2024) and transcripts from StringTie2 (Kovaka et al., 2019) as supporting evidence. This recovered 14 likely protein-coding loci that the annotation lacks. They are provided as `protein_match` features in version 3 of the annotation and are not counted among the 89 gene models of the inversion. Fixed differences were sites with an allele-frequency difference of at least 0.9 between 2a/2a and 2+/2+ homozygotes. Gene-ontology enrichment of the inversion genes used Fisher's exact test against the annotated-gene background in all three ontologies, compared with 500 random gene sets of equal size, and gene density was compared with all autosomal windows of the same length (0.5-Mb steps). The zero-fold fraction among the top 1% of coding sites by per-site $F_{\mathrm{ST}}$ (genotype likelihoods, 40 individuals per karyotype) was compared with all coding sites of the same region by Fisher's exact test, in the inversion ($p = 1 \times 10^{-32}$) and in collinear chr2:10–30 Mb ($p = 0.007$). Per-transcript $d_{\mathrm{xy}}$, the mean between-arrangement divergence over a transcript's coding sites (transcripts with at least 50), was normalised to its 100-kb window and compared with the same ratio for collinear control transcripts.

**Inversion coalescent model.** The inversion was simulated with msinv v0.4.1 (`https://github.com/stsmall/msinv`), a coalescent simulator with recombination that writes tree sequences. It uses the sequentially Markov approximation (McVean and Cardin, 2005), extended to two karyotype classes as in Peischl et al. (2013), in which two lineages coalesce only where they share ancestral material. The model code, inference scripts and every result quoted here are in the `illex/` and `results/illex/` directories of that repository (branch `illex/baseline-refit-karyofix`). Inside the inversion every lineage carries a karyotype label, ancestral (S, the 2+ arrangement) or derived (I, the 2a arrangement). Two lineages of the same class coalesce at rate $1/[2N(t)\,p_c(t)]$ per generation, where $N(t)$ is the effective size and $p_c(t)$ the frequency of their class, and a lineage of class $c$ recombines within the inversion at rate $r\,p_c(t)$, because crossing over occurs only between chromosomes of the same arrangement. Lineages of different classes cannot coalesce while the barrier exists, except through gene flux, which copies tracts of mean length 300 bp between arrangements, within the 188–390 bp estimated by Korunes and Noor (2019), with the position-dependent initiation rate of Peischl et al. (2013), rising from zero at the breakpoints toward the centre of the inversion. The gene-flux parameter is the per-site conversion probability per generation in heterokaryotypes. Backward in time, the barrier is removed at $t_{\mathrm{inv}}$, when the derived lineages rejoin the ancestral class. Each proposal simulated 40 kb of inversion between two 10 kb collinear margins, and a separate 40 kb of collinear sequence with msprime 1.4.1 (Baumdicker et al., 2022) as the control,

with 100 chromosomes of each arrangement and $\mu = 3 \times 10^{-9}$. For linkage disequilibrium the chromosomes of each arrangement were paired at random into 50 diploids and passed through the genotype error model of the data, in which a heterozygote is called homozygous with probability $h$ and each genotype is missing with probability one minus the call rate, both estimated for each arrangement and region from the same individuals as the observed statistics (call rates 0.62–0.67, $h = 0.64$–$0.69$).

**Neutral baseline.** The demography and recombination rate of the simulations were fitted to collinear chromosome-2 sequence flanking the inversion, chr2:40–55 and 85–115 Mb, by rejection ABC. This sequence lies 5.5–35.5 Mb outside the breakpoints and avoids a stretch of doubled diversity at chr2:24.1–35.9 Mb. The 2a/2a and 2+/2+ individuals were treated as two random sample sets, and the fitted statistics were the folded site-frequency spectrum in six bins, linkage disequilibrium in six distance bins, Tajima's $D$ and $\log_{10}\pi$, each averaged over the two sample sets, together with eight cells of the joint spectrum between them (the four cells for variants near fixation in one set and absent from the other are empty in collinear sequence and were left out). Diversity in this sequence, from the summed within-group ANGSD spectra, is $\pi = 0.0061$, against 0.0093 for the autosomes. The model was exponential growth from $N_{\mathrm{anc}}$ to $N_0$ over $T_{\mathrm{grow}}$ generations, constant before that, with recombination rate $r$ and $\mu$ fixed at $3\times10^{-9}$. 20,000 draws from log-uniform priors ($N_{\mathrm{anc}}$ $10^5$–$3 \times 10^6$, growth fold 1–200, $T_{\mathrm{grow}}$ $10^4$–$3 \times 10^6$, $r$ $10^{-10}$–$10^{-8}$) were each simulated for 200 kb with msprime and the control's genotype error model. The 1% closest to the data in a MAD-scaled Euclidean distance were accepted with Epanechnikov weights, and the log parameters were adjusted by local-linear regression (Beaumont et al., 2002). The posterior medians (95% intervals) are $N_{\mathrm{anc}} = 507{,}288$ (378,143–635,672), $N_0 = 9{,}112{,}696$ (7,632,032–11,683,078), $T_{\mathrm{grow}} = 382{,}957$ (314,983–472,238) and $r = 5.51 \times 10^{-10}$ ($4.61$–$6.49 \times 10^{-10}$). With $\mu$ fixed, only $r/\mu$ is identified, so $r$ is the effective rate for these statistics. The accepted draws reproduce the spectrum, Tajima's $D$, $\pi$, linkage disequilibrium from 240 bp to 20 kb, and the symmetric cells of the joint spectrum. Linkage disequilibrium at 100–240 bp lies above every accepted draw, and 2+/2+ carries more private rare variants than 2a/2a (0.352 against 0.272 of variable sites), which a single panmictic population does not produce. The inversion model used the posterior medians.

**Trajectory models and priors.** The frequency of 2a followed a Wright–Fisher diffusion in steps of 10 generations, with the effective size of the neutral baseline. Under neutral drift (N) there is no selection. Under directional selection the frequency changes deterministically by $s\,p(1-p)$ per generation from $t_{\mathrm{sel}}$ to the present (D1) or only until $t_{\mathrm{stop}}$, with drift alone after that (D2). Under balancing selection (B) it changes by $s\,p(1-p)(p^*-p)/p^*$ from $t_{\mathrm{sel}}$, the form expected under overdominance with equilibrium $p^*$. A hard start places a single copy of 2a at $t_{\mathrm{inv}}$, with selection from $t_{\mathrm{inv}}$, and its path is weighted by the exact probability that the copy establishes. A soft start places 2a at a founding frequency $p_0$ at $t_{\mathrm{inv}}$, with neutral drift until $t_{\mathrm{sel}}$. A path is kept only if 2a is neither lost nor fixed and ends within 0.02 of its present frequency of 0.602. The priors were $t_{\mathrm{inv}}$ uniform on $10^5$–$1.5 \times 10^6$ generations, $p_0$ log-uniform on $10^{-3}$–0.10, $t_{\mathrm{sel}} = f\,t_{\mathrm{inv}}$ and $t_{\mathrm{stop}} = f'\,t_{\mathrm{sel}}$ with $f$ and $f'$ uniform on 0–1, $s$ log-uniform on $10^{-6}$–$10^{-1}$, $p^*$ uniform on 0.3–0.95, the gene-flux rate log-uniform on $10^{-9}$–$10^{-5}$ when gene flux was fitted and $10^{-15}$ otherwise, and equal prior probabilities of a hard and a soft start. The upper bound of 0.10 on $p_0$, a sixth of the present frequency of 2a, keeps a soft start an arrangement that is still rare when the barrier forms, so that its rise to 0.602 is inferred rather than built into the start. Because the posterior of $p_0$ reaches this bound (fig. S10), the founding frequency is constrained by the prior above about 0.05.

**Summary statistics.** Thirty-nine statistics were fitted. Three are ratios between arrangements, $\pi_I/\pi_S$, $d_{xy}/\pi_I$ and the ratio of their singleton-class fractions. Twelve are the folded site-frequency spectra of the two arrangements projected to 20 chromosomes, as proportions of segregating sites in bins 1 to 5 and 6 or more. Twelve are the mean genotype $r^2$ between SNPs with a minor-allele frequency of at least 0.05 within each arrangement, in six log-spaced distance bins from 100 bp to 20 kb. Because unphased genotype $r^2$ depends on genotype error, the simulated genotypes carry the same heterozygote-dropout factor $h$ and call rate as the data, estimated per arrangement from the observed ratio of observed to expected heterozygosity ($h = 0.68$ in both arrangements). Twelve are the cells of the folded joint spectrum, with both arrangements projected to 20 chromosomes and binned at 0, 1–4, 5–15, 16–19 and 20 derived copies. Every statistic except the joint spectrum is divided by the same statistic in the control, in the simulations and in the data alike. In the data the ratios come from called genotypes of the 223 2a/2a and 97 2+/2+ individuals (pg_gpu), the spectra from per-arrangement ANGSD site-frequency spectra, the joint spectrum from a two-dimensional realSFS spectrum of 40 individuals of each homokaryotype (the first 40 of each karyotype list, with four 2a/2a individuals re-karyotyped by the depth-equalised call replaced by random 2a/2a individuals, seed 20261111; the cap keeps the two-dimensional estimation tractable), and linkage disequilibrium from 20 random subsamples of 50 individuals of each arrangement. The control is the collinear sequence of the neutral baseline, and its joint spectrum was built with the same 80 individuals, sites and filters as the inversion's. Three further statistics were withheld from the fit and used only to check it, the Tajima's $D$ of each arrangement and the ratio of $d_{xy}/\pi$ in windows near the breakpoints to that in windows at the centre of the inversion.

**ABC random-forest model choice.** The four histories were compared by ABC-RF (Pudlo et al., 2016) in abcrf, on a reference table simulated from the priors of the four histories with gene flux, with hard and soft starts simulated separately and each path kept when it passed the present-frequency filter (16,959 simulations after 19 with undefined statistics were removed, 3,228 for N, 3,744 for D1, 4,994 for D2 and 4,993 for B). Hard starts were weighted by their establishment probability, and each start class by its prior probability and its acceptance rate, so that each history carries its prior-weighted mass. A forest of 1,000 trees was grown on the 39 statistics and their linear discriminant axes, and the posterior probability of each history is the prior-weighted mass of the training simulations that share leaves with the observed statistics, multiplied by the probability that a prior draw of the history passes the present-frequency filter (0.0064 for B and 0.0005–0.0006 for each of the others). At the observed statistics the forest alone gives 0.31 for D2, 0.29 for B, 0.24 for N and 0.17 for D1, and with the acceptance rates the posterior probabilities are 0.83 for B, 0.08 for D2, 0.06 for N and 0.04 for D1. The classifier was assessed by grouped five-fold cross-validation (Supplementary Fig. S9). Its accuracy is 0.51 against 0.25 by chance. Neutral drift and ongoing directional selection are recovered best (0.86 and 0.68), whereas balancing selection is recovered in 0.36 of its mass and confused mainly with neutral drift (0.33), and a stalled sweep is recovered in 0.14 and confused with the other three.

**Simulated length.** Each simulation covers 40 kb of inversion, against 18.96 Mb in the data. To test whether this length represents the whole inversion, 48 NPE posterior draws of the balancing model with gene flux were each given one frequency path, and the same path was simulated over 40 kb, 160 kb, 640 kb, 2.56 Mb and the full 18.96 Mb. For the 47 draws with a valid path (one draw produced none within 200 attempts), the mean of every fitted and withheld statistic at 18.96 Mb lies within 0.2 standard deviations (of the 40-kb values across draws) of its mean at 40 kb. The spread across draws is smaller at 18.96 Mb, 0.3–1.0 of that at 40 kb for the fitted statistics and

0.12 for the edge-to-centre ratio. The 40-kb simulations therefore reproduce the expected statistics of the full inversion with more variance, which widens the posterior rather than shifting it.

**Neural posterior estimation.** Parameters of the balancing model were estimated by neural posterior estimation (Cranmer et al., 2020; Min et al., 2026) in sbi 0.27 (Tejero-Cantero et al., 2020; Boelts et al., 2025). The training simulations were new draws from the prior of the balancing model with a founding frequency, with and without gene flux, kept when the present frequency of 2a fell within 0.02 of 0.602, with the same simulator, baseline and 39 statistics (59,743 simulations with gene flux and 59,531 without, each from about 5.4 million prior draws). Starts from a single copy, which carry 3% of the posterior mass of balancing selection in the ABC-RF reference table, were not included. The parameters were transformed to the scale on which their priors are uniform ($t_{\mathrm{inv}}$, $t_{\mathrm{sel}}/t_{\mathrm{inv}}$, $\log_{10} p_0$, $\log_{10} s$, $p^*$ and the log gene-flux rate), and the statistics were standardised. The posterior is an equal mixture of five neural spline flows (Durkan et al., 2019) with five transforms and 64 hidden units, each trained from a different seed with 10% of the training set held for validation and stopped after 30 epochs without improvement. Calibration was tested on 2,000 held-out simulations by simulation-based calibration (Talts et al., 2018) and by the TARP coverage test (Lemos et al., 2023). With gene flux the central 50, 90 and 95% intervals contain the true value in 49–52, 89–91 and 94–96% of simulations, and the rank distributions are uniform (Kolmogorov–Smirnov $p$ 0.12–0.68) except for a slight deviation for $p^*$ ($p = 0.02$). The observed statistics lie inside the training distribution, at the 0.27 quantile of the robust Mahalanobis distance and between the 29th and 89th percentile for each statistic. Without gene flux the coverage is also nominal and the ranks are uniform ($p$ 0.09–0.64), but about half of the training simulations have no variants shared at low frequency by the two arrangements, which the data have. Twenty thousand posterior draws were taken at the observed statistics. The trajectories in main-text Fig. 4b and Supplementary Fig. S10 were simulated from 400 posterior draws with gene flux, each re-simulated until its path passed the present-frequency filter (392 within 50 attempts).

**Posterior predictive checks.** The balancing model was checked from 1,000 NPE posterior draws for each gene-flux setting. Each draw was simulated again with the same simulator, with the frequency path retried until it passed the present-frequency filter. Each statistic was scored by the quantile of the observed value among the replicates and a two-sided predictive $p$-value, and the joint fit by the rank of the observed distance from the predictive median, in the distance of the fit, among the replicates' distances (Supplementary Fig. S11 for the balancing model with gene flux). Under balancing selection with gene flux the joint predictive $p$ is 0.51 (0.09 without gene flux). The withheld statistics are reproduced (two-sided $p$ 0.84 and 0.56 for Tajima's $D$ in 2a and 2+, 0.99 for the edge-to-centre ratio). With gene flux, no statistic falls below $p = 0.05$, and the cell of variants at low frequency in both arrangements, linkage disequilibrium at 8–20 kb in 2+ and at 100–580 bp in 2a, and the cell of variants rare in 2a and fixed in 2+ are at the edge of their ranges ($p = 0.07$–$0.14$). Without gene flux, no statistic falls below $p = 0.05$ either, and linkage disequilibrium at 8–20 kb in 2+ ($p = 0.07$) and the cells of variants rare in 2a and at high frequency or fixed in 2+ are at the edge of their ranges (observed 0.016 and 0.011, predicted 0.002 and 0.0002, $p = 0.12$ and 0.06). The cell of variants at low frequency in both arrangements is reproduced in both settings (observed 0.064, predicted 0.031 with gene flux and 0.035 without, $p = 0.07$ and 0.29). In the collinear flanks the same cell matches the neutral baseline (observed 0.233, accepted draws 0.210–0.269), so the two-dimensional spectrum and the baseline reproduce it where there is no barrier.

**Recombination map.** ReLERNN (Adrion et al., 2020) was run in unphased mode on biallelic SNPs with at most 50% missing genotypes. Individuals with no genotype calls on a chromosome cannot enter the unphased model, and the 329 confidently sexed males with data were used. The autosomal network was trained on chromosomes 1, 13 and 45 with 12,000 coalescent simulations (with 1,000 each for validation and testing) under the Stairway Plot 2 demography with $\mu = 3 \times 10^{-9}$, an upper bound of 4 on the ratio of $\rho$ to $\theta$, and 6-kb windows, masked by the inaccessible regions and the 22 $\pi$-elevated blocks with a window-level masking threshold of 0.9. It was trained for 50 epochs and applied to every autosome except chromosome 2, in windows of 3–6 kb. The Z network was trained in the same way in 30-kb windows, with the Z accessibility mask and the effective sizes of the demography multiplied by 0.60, the Z-to-autosome diversity ratio. Predictions were corrected with 200 parametric-bootstrap replicates in each of 20 rate slices. The map pools all karyotypes and reads linkage disequilibrium within windows of a few kilobases, so it does not resolve the barrier between chromosome-2 arrangements, which is shown instead by long-range linkage disequilibrium (Supplementary Figs. S8, S19).

**Z recombination and sex-biased mutation.** Linkage disequilibrium on the Z and on the autosomes was compared in the 329 ZZ males, the only individuals that carry two Z copies. Two-locus statistics ($D^2$ and $\pi_2$, unbiased for unphased genotypes (Ragsdale and Gravel, 2020)) were computed with pg_gpu (Pope et al., 2026) on accessible biallelic SNPs in 2-Mb windows, 86 on the autosomes (at 10 and 40 Mb of each autosome, excluding the chromosome-2 inversion and the anomalous-diversity blocks) and nine spaced every 4 Mb along the Z, with SNP pairs binned by distance from 0 to 5 kb. The decay of $\sigma_D^2 = D^2/\pi_2$ with distance was fitted to the expectation of Ohta and Kimura (1971) plus a constant sample-size term, giving the population-scaled rate $\rho = 4N_e r$ per base for each chromosome class. The ratio $\rho_Z/\rho_A$ was also estimated without a model, as the distance rescaling that best superimposes the two decay curves. Confidence intervals come from 200 bootstrap resamples of windows. Because the Z recombines only in males and spends two thirds of its time in them, $\rho_Z/\rho_A = (N_{e,Z}/N_{e,A})\,(4/3)\,r_m/(r_m + r_f)$, where $r_m$ and $r_f$ are the male and female per-generation rates and the male rate is taken to be the same on the Z and the autosomes. With $N_{e,Z}/N_{e,A} = 3/4$, equal recombination in the two sexes predicts $\rho_Z/\rho_A = 0.5$ and achiasmate female meiosis predicts 1 (Coffing et al., 2026).

Sex-biased mutation was estimated from synonymous divergence between *Illex illecebrosus* and *S. oualaniensis*, whose chromosome 46 is syntenic with the *Illex illecebrosus* Z (Small et al., 2026). The *Illex illecebrosus* annotation was transferred to the *S. oualaniensis* assembly with LiftOn (Chao et al., 2025), one transcript per gene was aligned codon by codon with MACSE 2.07 (Ranwez et al., 2018), which allows frameshifts in the transferred sequence, and codons with gaps, frameshifts, ambiguous bases or stop codons were removed. Synonymous and nonsynonymous divergence ($d_S$, $d_N$) were computed per gene by the method of Nei and Gojobori (1986) with a Jukes–Cantor correction. Genes in the chromosome-2 inversion or the anomalous-diversity blocks were excluded, leaving 8,649 autosomal and 109 Z-linked genes with $d_S > 0$. The male-to-female mutation ratio $\alpha = \mu_m/\mu_f$ was obtained from the ratio $R$ of mean Z to mean autosomal $d_S$ as $\alpha = (3R-2)/(4-3R)$, following Miyata et al. (1987), as in Coffing et al. (2025) and Coffing et al. (2026). Z-linked genes carry more gene-body CpG methylation than autosomal genes in *Illex illecebrosus*, most strongly in females (Small et al., 2026). If this extends to the germline, methylation-associated mutation raises the Z rate and biases $\alpha$ upward, so the female bias is if anything underestimated. To separate mutation from selective constraint and base composition, $R$ was recomputed with autosomal genes matched to the Z genes by deciles of $d_N/d_S$ and of third-position GC content (GC3), and with genes restricted to those above the autosomal median $d_N/d_S$. Confidence intervals come from

2,000 bootstrap resamples of genes within strata.

Linkage disequilibrium decays at the same rate on the Z as on the autosomes, $\rho_Z/\rho_A = 1.04$ (95% CI 0.88–1.28) from the decay fit and 1.08 (0.78–1.67) from curve rescaling, with $\rho_A = 0.026$ per base (Supplementary Table S9). Synonymous divergence is lower on the Z ($d_S = 0.298$ vs 0.358, Mann–Whitney $p = 0.009$), giving $R = 0.83$ and $\alpha = 0.33$ (95% CI 0.10–0.70), a female mutation rate about three times the male rate. Z-linked genes are slightly more constrained (median $d_N/d_S$ 0.082 vs 0.113, $p = 0.048$) and lower in GC3 (0.36 vs 0.42, $p < 10^{-16}$). Matching on $d_N/d_S$ leaves $\alpha$ unchanged (0.36), and matching on GC3 raises it to 0.43, or 0.49 when combined with $d_N/d_S$, so the female bias holds under each control, with the upper confidence limit reaching 1 only under joint matching. Restricting to genes above the autosomal median $d_N/d_S$ leaves only 44 Z genes and gives $\alpha = 0.24$ (0–0.65), in line with the other estimates but less precise. Because $R$ estimates $\mu_Z/\mu_A$, the ratio of male Z to autosomal diversity ($\pi_Z/\pi_A = 0.00563/0.0093 = 0.61$) implies $N_{e,Z}/N_{e,A} = 0.68$–$0.76$ across these estimates, close to the 3/4 expected under an equal breeding sex ratio. Combined with the linkage-disequilibrium ratio, this gives $r_m/(r_m+r_f) \approx 1.02$–1.14, consistent with achiasmate female meiosis (Coffing et al., 2026) and inconsistent with equal recombination in the two sexes, which predicts 0.5.

**Selective-sweep scan.** diploS/HIC used 1.1-Mb feature windows of eleven 100-kb sub-windows and twelve diploid statistics per sub-window. It was trained on hard and soft sweeps simulated in SLiM 5.2 and recapitated with msprime, and on neutral msprime simulations, all under the fitted growth demography, at the empirical sample size, and with the empirical per-individual missing-data structure. The final model classified held-out simulations with 81% accuracy (Supplementary Fig. S20). Applied to the genome it assigns 1% of windows to the neutral class and 91% to the linked classes, because real neutral windows carry three to eight times more spatial heterogeneity in diversity among sub-windows than the simulations. We therefore used it as an outlier detector. The sweep score $p_{\text{hard}} + p_{\text{soft}}$ was ranked within strata of callable fraction (quintiles) and coding density (zero or not), and windows in the top 1% of their stratum were outliers. RAiSD and SweepFinder2 statistics were ranked the same way, and a region was concordant when a diploS/HIC outlier was supported by either. The false-discovery rate of the concordant set was estimated by shifting the joint RAiSD and SweepFinder2 outlier track circularly within each chromosome by an independent uniform offset of whole 100-kb windows, 10,000 times, and counting the diploS/HIC outlier regions that became concordant by chance (227 outlier regions, 446 footprint outlier windows among 22,278 callable windows). The mean of this null, 4.8 regions (95% range 1–9), against the 24 observed gives a false-discovery rate of 0.20 (no shift reached 24). Tier-1 calls are stratified outliers by construction and have no comparable estimate. Tier-1 calls came from a hidden Markov model with neutral, linked and sweep-centre states, whose emissions are the prior-corrected class posteriors and whose dwell lengths follow the ∼400-kb sweep footprint scaled by the recombination map. The decoder is not calibrated by simulation, because at this population size the simulated ancestral recombination graphs carry about $3.4 \times 10^5$ marginal trees per 2.5 Mb. Its settings are design-based and were checked against the concordant regions (Supplementary Fig. S22). The background-selection screen compared each call's $\pi$, from the ANGSD windows it overlaps, with the value predicted from the fraction of coding sites within 1 Mb of its midpoint, $\hat{\pi} = 0.010593 - 0.165473\, f_{\text{CDS}}$, the genome-wide linear relationship between the two. A call was retained when $\pi - \hat{\pi} \leq -0.003$, i.e. when its diversity is at least 0.003 below the background-selection expectation. Retained calls with $f_{\text{CDS}}$ above the genome median are those well below the expectation, and those below it are low-diversity calls in gene-poor sequence where background selection is weak. The target of each of the 11 retained autosomal candidates was localised with the ANGSD 10-

kb windows (`pipeline/steps/14_sweep_seqmodel/scripts/harness/auto_finescan.py`). The window of lowest $\pi$ within 0.5 Mb of the call's midpoint was taken as the centre of the sweep, and the annotated gene within 7.5 kb of it as the target. All 11 centres lie in the lowest 6% of their chromosome's $\pi$, with Tajima's $D$ of $-2.70$ to $-2.42$. Three fall inside the 100-kb call and eight 65–475 kb beside it, and two of these move the target to a neighbouring gene (*Cpes* at chr1:85.5 Mb, about 11 kb from a centre in a gene desert, and *GALT* at chr25:65.0 Mb). The Z candidates were localised in the same way from 10-kb windows of $\pi$ and Tajima's $D$ computed from the called genotypes of the males (`chrZ_finescan.py`). Gene-ontology enrichment used genes with EnTAP (Hart et al., 2020) annotations that overlap a call, against all annotated genes outside the Z and the masked blocks.

**Anomalous-diversity blocks.** Blocks were called where the rolling median of 10-kb $\pi$ exceeded twice the chromosome median over at least 1 Mb. Each was characterised by per-division $\pi$, between-division $F_{ST}$, local PCA, and genotype linkage disequilibrium, and by read depth relative to 5-Mb flanks in all 621 individuals (samtools bedcov at mapping quality 0 and 20). Long-read depth and heterozygous-site allele fractions came from PacBio HiFi reads of the assembly individual and a second individual aligned to the assembly, and genotype-likelihood $\pi$ was recomputed at mapping quality 60. Blocks with population depth at least 1.15 times their flanks were classed as collapsed duplications and those at 0.90 or less as under-mapped, and the fraction of each block's accessible sequence that is not uniquely mappable was taken from GenMap (Supplementary Table S5).

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## Supplementary figures

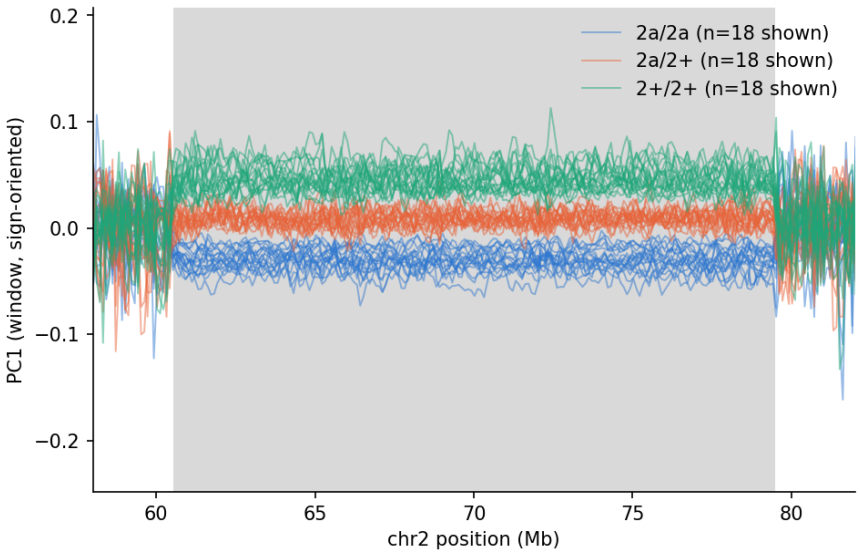



Figure S1: **Windowed PCA across chromosome 2** localising the differentiated block.

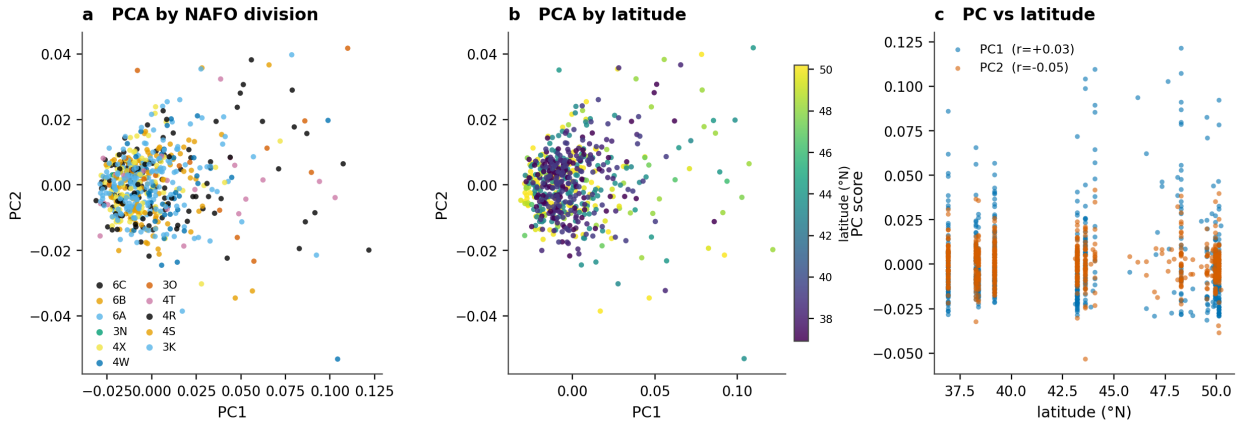



Figure S2: **The autosomal-PCA spread is not geographic structure.** Genome-wide PCA (all autosomes except chromosome 2, 280,131 LD-thinned SNPs), $n = 619$ individuals. (a) Coloured by NAFO sampling division and (b) by latitude, the divisions and the latitudinal gradient are fully intermixed across PC space, with no clustering by origin. (c) PC scores plotted against latitude show no relationship (Pearson $r = 0.03$ for PC1, $r = -0.05$ for PC2). PC1 explains 1.4% of the variance and tracks sequencing depth (Spearman $\rho = -0.99$ with the number of autosomal reads). PC2 explains 0.4% and is carried by four individuals (80% of its variance).

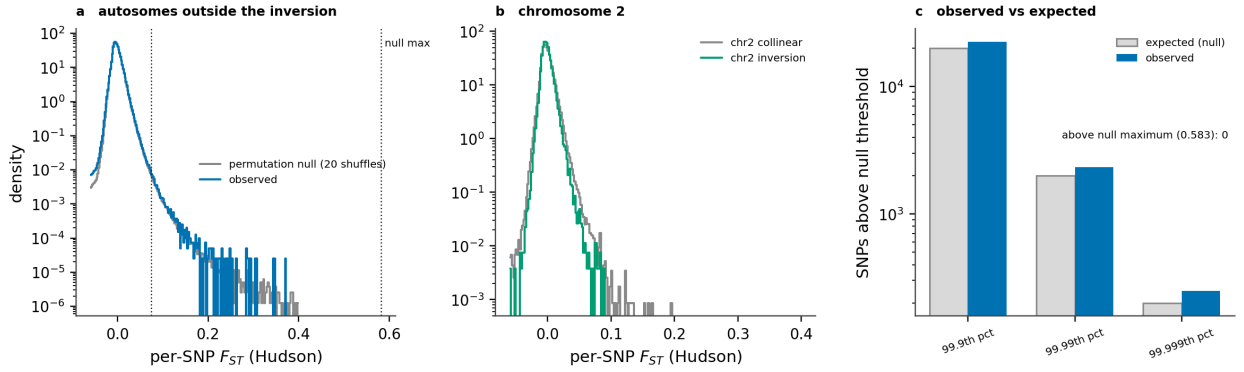



Figure S3: **No geographic differentiation beyond genetic drift and no local adaptation signal.** Per-SNP Hudson $F_{\mathrm{ST}}$ across all 45 NAFO-division pairs for every biallelic SNP with minor-allele frequency above 0.05 ($\geq$40 genotyped individuals per pair, ratio of averages over $\geq$20 informative pairs). **(a)** On the autosomes outside the inversion (19,984,027 SNPs), the empirical distribution (blue) matches the permutation null (grey), built by shuffling division labels among individuals 20 times, which keeps genotype calling, missingness, the allele-frequency spectrum and linkage. Empirical maximum $F_{\mathrm{ST}} = 0.371$ against a null maximum of 0.583. **(b)** On chromosome 2, the inverted region (green) is no more geographically differentiated than the collinear remainder, so the inversion is non-clinal and its $F_{\mathrm{ST}} \approx 0.36$ is between karyotypes, not geography. Its upper tail is lighter than the null (66 SNPs above the null 99.9th percentile against 134 expected). **(c)** SNPs above the null 99.9th, 99.99th and 99.999th percentiles (22,562, 2,319 and 249) are within the permutation range ($19{,}979 \pm 2{,}154$, $1{,}998 \pm 352$ and $200 \pm 54$, permutation $p = 0.19$, 0.24 and 0.29), and no SNP exceeds the null maximum, so a scan for local adaptation that the diversity-based sweep tests could miss returns no candidates. Because the minor-allele-frequency filter is applied to the pooled 620 individuals, a variant private to one small division can fall below it (a SNP fixed only in 3O, with 16 individuals, has a pooled frequency of 2.5%), so the scan rules out strongly differentiated SNPs that are common overall, but not private alleles in the smallest divisions.

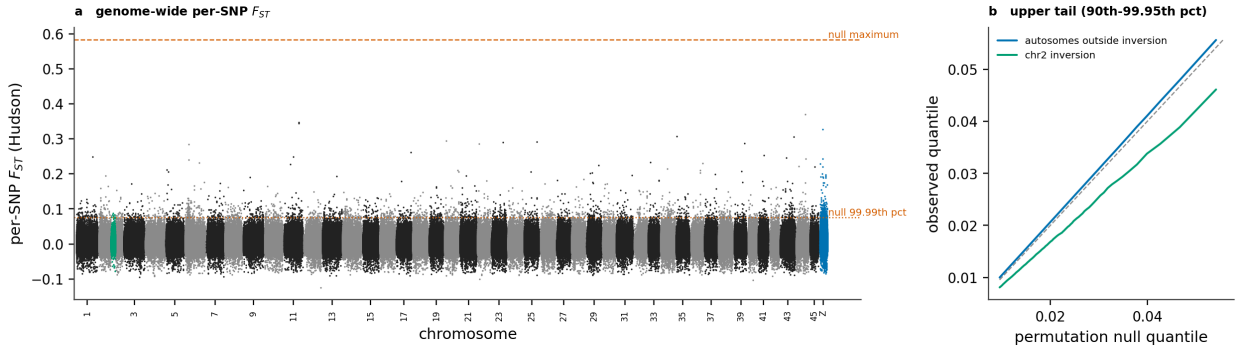



Figure S4: **Genome-wide per-SNP $F_{ST}$ finds no differentiation outlier.** Per-SNP Hudson $F_{ST}$ of fig. S3 along the genome (20,276,930 SNPs with minor-allele frequency above 0.05 that are informative in at least 20 division pairs, 20,117,984 on the autosomes and 158,946 on the Z, every SNP shown). **(a)** No SNP exceeds the permutation-null maximum (0.583, dashed). The 2,327 SNPs above the null 99.99th percentile (0.075, dotted) on the autosomes match the number expected by chance and are spread across all chromosomes. The chromosome-2 inversion is green. chrZ (blue) exceeds the null tail (187 SNPs above the 99.99th percentile against 16 expected) because division sex ratios differ and females are hemizygous, and it is shown for completeness. **(b)** Upper-tail quantiles of the observed against the permutation null. The autosomes outside the inversion follow the identity line, and the inversion lies below it.

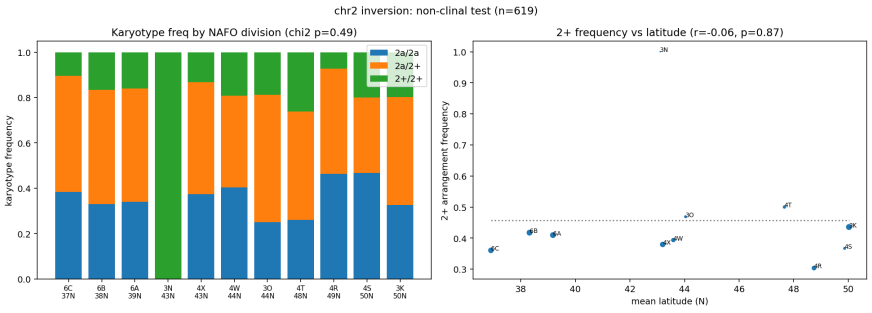



Figure S5: **The inversion is non-clinal.** Frequency of the ancestral 2+ arrangement across NAFO sampling divisions shows no latitudinal trend (see table S2). Geographic $F_{ST} \approx 0$, and the three karyotypes co-occur throughout the range.

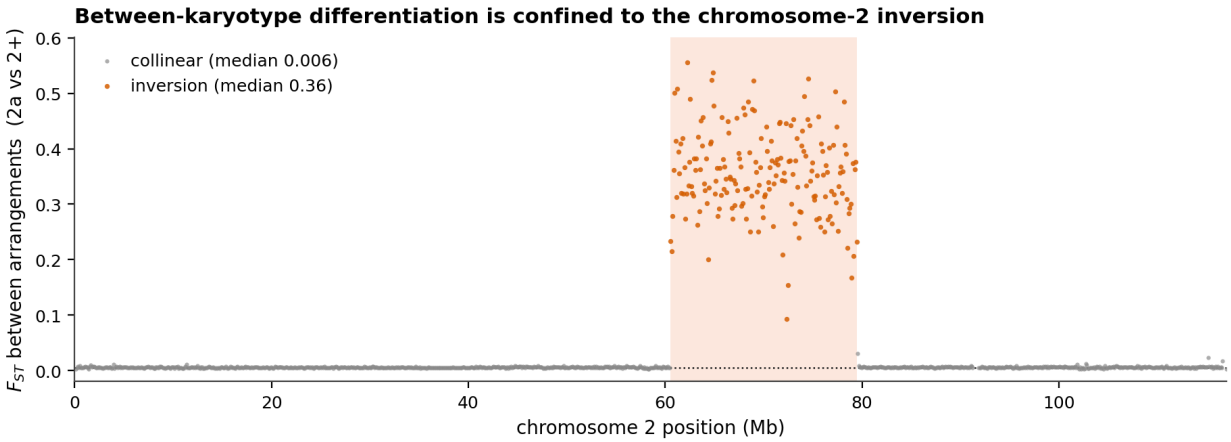



Figure S6: **Between-karyotype differentiation is confined to the chromosome-2 inversion.** Per-window $F_{\mathrm{ST}}$ between the 2a and 2+ arrangements (arrangement homozygotes, 2a/2a $n = 223$ and 2+/2+ $n = 97$) across the whole of chromosome 2, aggregated in 100 kb windows (ratio-of-averages Hudson estimator). Unlike the population-$F_{\mathrm{ST}}$ scan (which asks about geography), this asks how differentiated the two arrangements are along the chromosome. The collinear remainder (grey) sits at $F_{\mathrm{ST}} \approx 0$ (median 0.006), so outside the inversion the karyotypes are one panmictic population, while the inversion (60.54–79.50 Mb, shaded) is strongly differentiated (median 0.36, max 0.56). The whole-inversion $F_{\mathrm{ST}}$ of 0.36 that defines the "structure" on chromosome 2 is thus between karyotypes and confined to the inversion, not geographic.

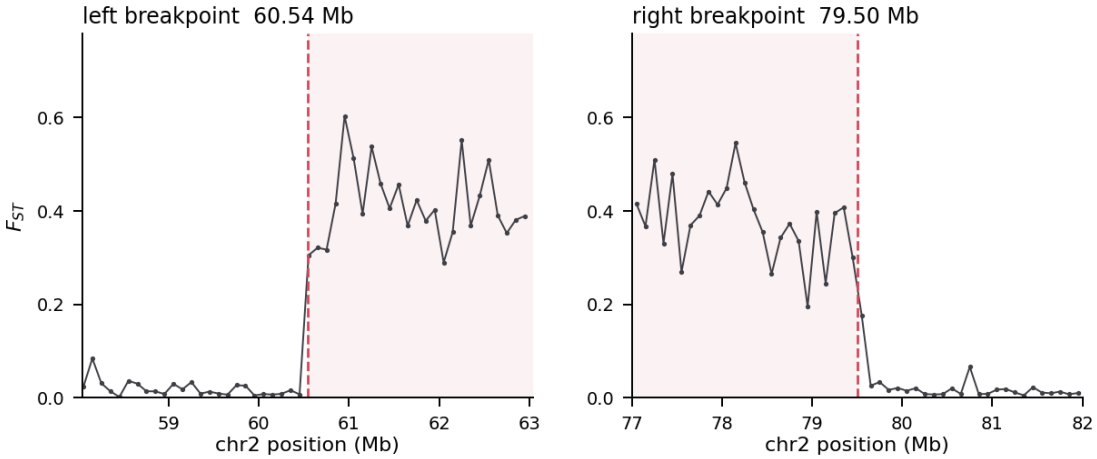



Figure S7: **Inversion breakpoints.** $F_{\mathrm{ST}}$ across the left (60.54 Mb) and right (79.50 Mb) breakpoints, a step rather than a ramp.

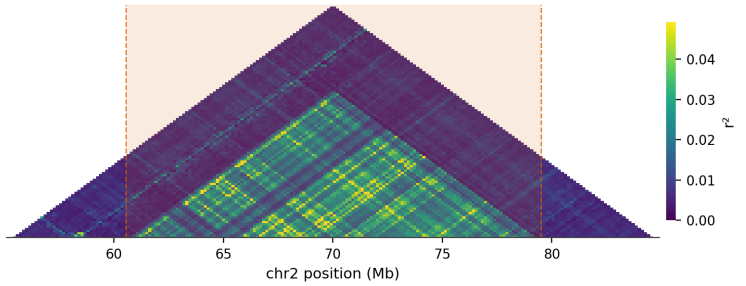



Figure S8: **Linkage disequilibrium across the chromosome-2 inversion.** Mean genotype $r^2$ between SNPs, binned along chr2:55–85 Mb, pooling all 621 individuals. Dashed lines and shading mark the breakpoints (60.54 and 79.50 Mb). A block of elevated linkage disequilibrium spans exactly the inversion and stops at both breakpoints, while the flanking collinear sequence shows only background values. Mean $r^2$ between SNPs more than 1 Mb apart is 3.5 times higher inside the inversion than in its flanks (0.0056 vs 0.0016, Supplementary Fig. S19). This long-range association arises because the few near-fixed differences between arrangements are all correlated with karyotype, and it is the direct signature of suppressed recombination between arrangements. Recombination within arrangements is not suppressed at the scale of the 3–6-kb windows used by the recombination map, which is why the inversion is shown here rather than in Supplementary Fig. S14.

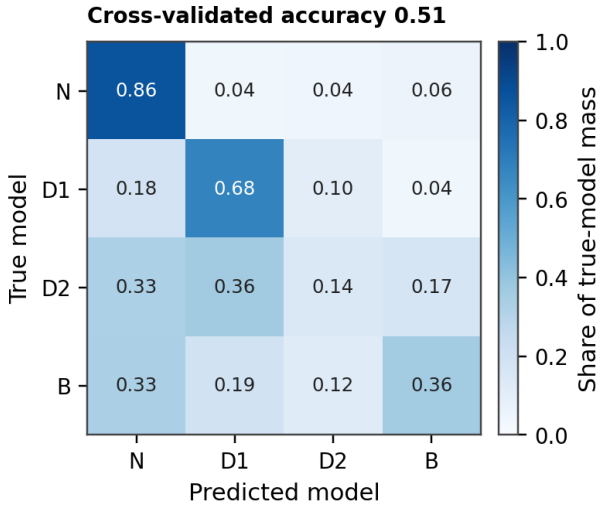



Figure S9: **Confusion matrix of the ABC-RF model choice for the chromosome-2 inversion.** Grouped five-fold cross-validation over 16,959 simulations of the four histories with gene flux, neutral drift (N), directional selection still under way (D1) or stopped (D2) and balancing selection (B). Each row is the true history, and the cells give the share of its prior-weighted mass assigned to each predicted history, so rows sum to 1. This is the classification by the forest alone, without the acceptance-rate weighting that gives $P(\mathrm{B}) = 0.83$ (Supplementary Methods).

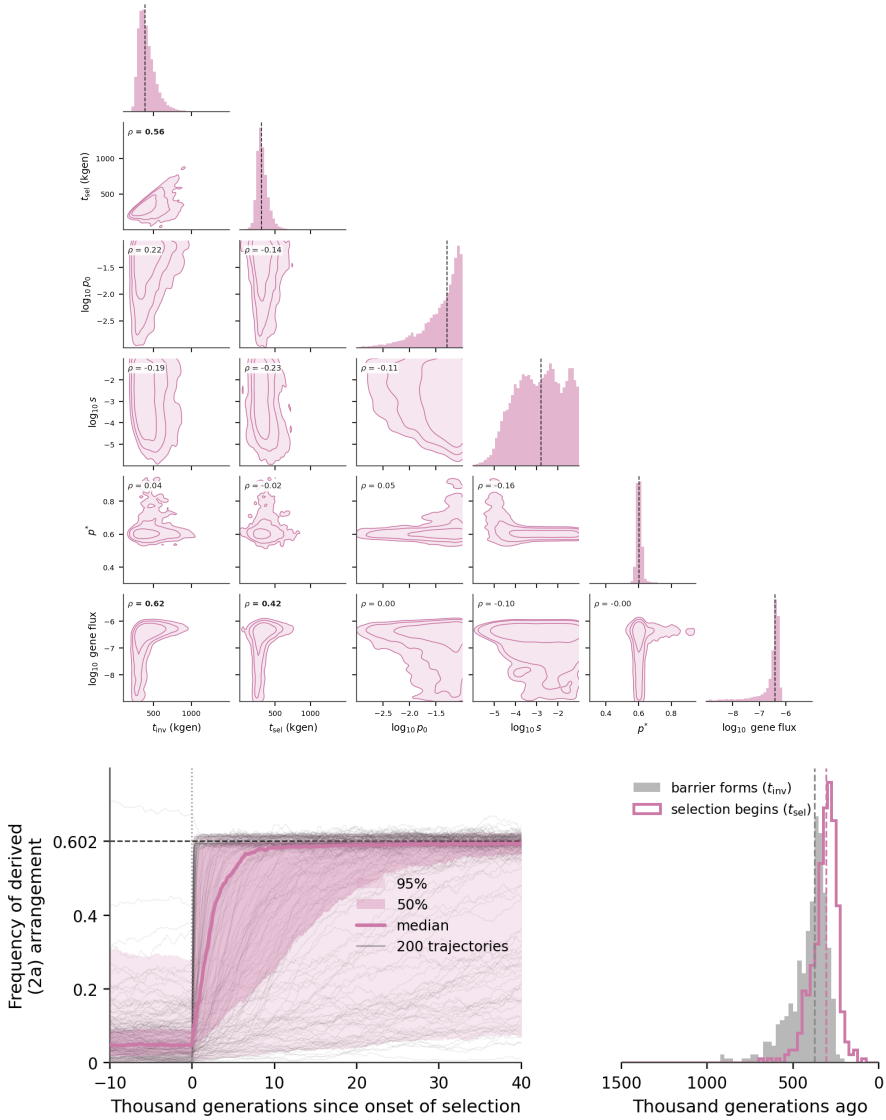



Figure S10: **Posterior of the balancing model with gene flux for the chromosome-2 inversion** (main-text Fig. 4b). Top, joint posterior from neural posterior estimation, with each axis spanning its prior. The diagonal gives the marginals with medians dashed, and below it the contours enclose 80, 95 and 99% of the posterior, with the Spearman correlation in each panel (bold where $|\rho| \geq 0.3$). The barrier age $t_{\text{inv}}$ and the onset of selection $t_{\text{sel}}$ are in thousands of generations. All draws are soft starts, which carry 97% of the posterior mass of balancing selection. The selection coefficient has only a lower bound and follows the prior above about $10^{-4}$. The gene-flux rate is the per-site conversion probability per generation in heterokaryotypes. The posterior of the founding frequency $p_0$ extends to the upper limit of its prior at 0.10. Bottom left, frequency of 2a in 200 trajectories re-simulated from posterior draws, aligned on the onset of selection, with bands (pointwise median, 50% and 95% ranges) computed from all 392 re-simulated trajectories. The median trajectory reaches 0.55 about 6,000 generations after onset, and the slowest fifth, mostly those with $s < 10^{-4}$, take more than 50,000 generations. Bottom right, the posterior draws of $t_{\text{inv}}$ and $t_{\text{sel}}$ behind these trajectories, with medians dashed.

## Supplementary tables

Table S1: **Sampling events.** Every collection event (NAFO division, date, depth) with its source, the number of individuals analysed in this study (618 of the 633 whole-genome sequenced individuals, excluding 12 whose BAMs contain essentially no data and three likely contaminated or mixed samples) and the number retained by Baker et al. (2025) after their genotype-missingness filter (540). Divisions are ordered north to south by mean sampling latitude. 4X was bulk-sampled across the division with no per-tow depth or position recorded.

| Division | Date | Source | Depth (m) | $N$ (this study) | $N$ (Baker) |
|---|---|---|---|---|---|
| 3K | 16 August 2023 | Recreational jig | 3 | 47 | 48 |
| | 30 August 2023 | Coastal fishery jig | 5 | 53 | 46 |
| 4S | 28 August 2023 | Canadian trawl survey | 228 | 2 | 2 |
| | 29 August 2023 | Canadian trawl survey | 251 | 1 | 1 |
| | 30 August 2023 | Canadian trawl survey | 80 | 1 | 1 |
| | 30 August 2023 | Canadian trawl survey | 124 | 2 | 2 |
| | 31 August 2023 | Canadian trawl survey | 152 | 2 | 1 |
| | 31 August 2023 | Canadian trawl survey | 155 | 3 | 3 |
| | 1 September 2023 | Canadian trawl survey | 273 | 1 | 1 |
| | 1 September 2023 | Canadian trawl survey | 296 | 1 | 1 |
| | 3 September 2023 | Canadian trawl survey | 243 | 2 | 2 |
| 4R | 7 August 2023 | Canadian trawl survey | 113 | 36 | 24 |
| | 25 August 2023 | Canadian trawl survey | 173 | 2 | 2 |
| | 25 August 2023 | Canadian trawl survey | 233 | 5 | 4 |
| | 25 August 2023 | Canadian trawl survey | 245 | 9 | 7 |
| | 28 August 2023 | Canadian trawl survey | 221 | 4 | 3 |
| 4T | 2 September 2023 | Canadian trawl survey | 243 | 1 | 0 |
| | 7 September 2023 | Canadian trawl survey | 27 | 1 | 1 |
| | 8 September 2023 | Canadian trawl survey | 49 | 1 | 0 |
| | 9 September 2023 | Canadian trawl survey | 112 | 1 | 0 |
| | 9 September 2023 | Canadian trawl survey | 183 | 1 | 0 |
| | 9 September 2023 | Canadian trawl survey | 306 | 1 | 0 |
| | 10 September 2023 | Canadian trawl survey | 81 | 1 | 1 |
| | 10 September 2023 | Canadian trawl survey | 101 | 1 | 1 |
| | 10 September 2023 | Canadian trawl survey | 285 | 1 | 0 |
| | 12 September 2023 | Canadian trawl survey | 285 | 1 | 0 |
| | 15 September 2023 | Canadian trawl survey | 50 | 1 | 1 |
| | 22 September 2023 | Canadian trawl survey | 39 | 1 | 1 |
| | 22 September 2023 | Canadian trawl survey | 142 | 2 | 1 |
| | 22 September 2023 | Canadian trawl survey | 158 | 1 | 1 |
| | 23 September 2023 | Canadian trawl survey | 341 | 2 | 2 |
| | 24 September 2023 | Canadian trawl survey | 118 | 1 | 0 |
| | 27 September 2023 | Canadian trawl survey | 36 | 1 | 0 |

| Division | Date | Source | Depth (m) | $N$ (this study) | $N$ (Baker) |
|---|---|---|---|---|---|
| | 28 September 2023 | Canadian trawl survey | 56 | 1 | 1 |
| | 28 September 2023 | Canadian trawl survey | 62 | 1 | 1 |
| | 29 September 2023 | Canadian trawl survey | 49 | 2 | 1 |
| 3O | 13 October 2023 | Canadian trawl survey | 204 | 14 | 7 |
| | 14 October 2023 | Canadian trawl survey | 350 | 1 | 1 |
| | 14 October 2023 | Canadian trawl survey | 613 | 1 | 1 |
| 4W | 16 July 2023 | Canadian trawl survey | 84 | 47 | 43 |
| 4X | July 2022 | Canadian trawl survey | not recorded | 83 | 76 |
| 3N | 20 October 2023 | Canadian trawl survey | 160 | 1 | 1 |
| 6A | 22 September 2022 | US trawl survey | 120 | 94 | 84 |
| 6B | 18 June 2023 | Commercial trawl | 74 | 49 | 46 |
| | 16 August 2023 | Commercial trawl | 201 | 48 | 48 |
| 6C | 14 September 2022 | US trawl survey | 307 | 86 | 73 |
| Total | | | | 618 | 540 |

Table S2: **Chromosome-2 karyotype counts and derived-arrangement frequency by NAFO division** (ordered by latitude). The derived (2a) frequency $p_{2a}$ shows no latitudinal cline (range 0.50–0.70, excluding 3N with one individual), consistent with geographic $F_{ST} \approx 0$ (fig. S5).

| Division | 2a/2a | 2a/2+ | 2+/2+ | $N$ | $p_{2a}$ | mean lat. (°N) |
|---|---|---|---|---|---|---|
| 6C | 33 | 44 | 9 | 86 | 0.640 | 36.9 |
| 6B | 32 | 49 | 16 | 97 | 0.582 | 38.3 |
| 6A | 32 | 47 | 15 | 94 | 0.590 | 39.2 |
| 3N | 0 | 0 | 1 | 1 | 0.000 | 43.1 |
| 4X | 31 | 41 | 11 | 83 | 0.620 | 43.2 |
| 4W | 19 | 19 | 9 | 47 | 0.606 | 43.6 |
| 3O | 4 | 9 | 3 | 16 | 0.531 | 44.0 |
| 4T | 6 | 11 | 6 | 23 | 0.500 | 47.7 |
| 4R | 26 | 26 | 4 | 56 | 0.696 | 48.7 |
| 4S | 7 | 5 | 3 | 15 | 0.633 | 49.9 |
| 3K | 32 | 48 | 20 | 100 | 0.560 | 50.0 |
| Total | 222 | 299 | 97 | 618 | 0.601 | – |

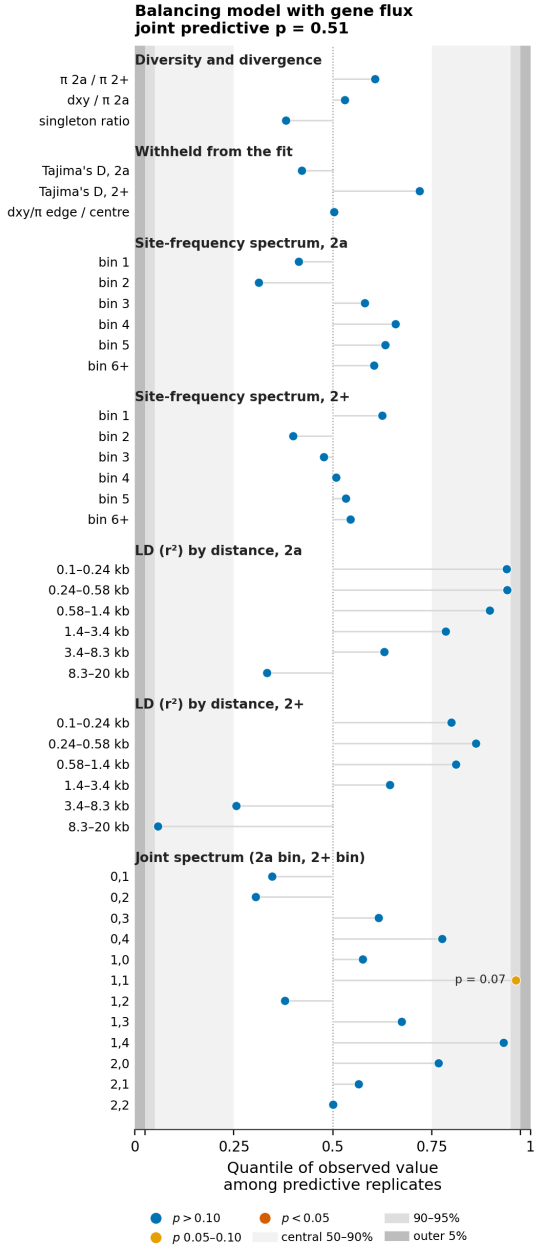



Figure S11: **Posterior predictive check of the balancing model with gene flux.** 1,000 NPE posterior draws were simulated again, as 40-kb segments of inversion like the training simulations, and each point is the quantile of the observed statistic among the replicates. The shading marks the central 50–90%, the 90–95% and the outer 5% of the predictive distribution, the points are coloured by the two-sided predictive $p$ (below 0.05, 0.05–0.10, above 0.10), and statistics with $p$ below 0.10 are labelled. The title gives the joint predictive $p$. Tajima's $D$ and the edge-to-centre divergence ratio were withheld from the fit. The site-frequency-spectrum bins are the proportions of segregating sites with 1 to 5 and 6 or more minor-allele copies of 20, and the joint-spectrum cells give the count bins of 2a and 2+ (0, 1–4, 5–15, 16–19 and 20 of 20). The check of the model without gene flux is summarised in the Supplementary Methods.

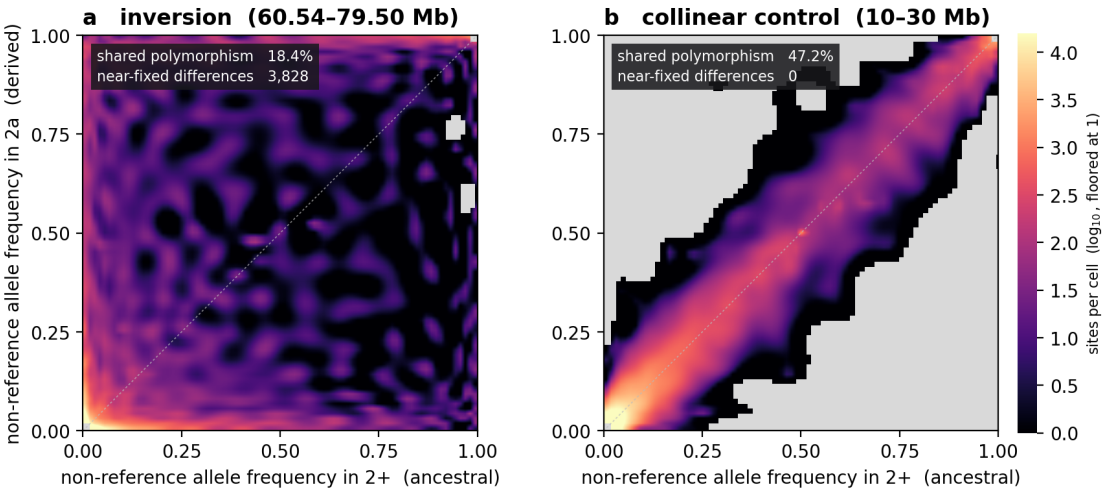



Figure S12: **Joint two-dimensional site-frequency spectrum** of non-reference allele frequency in the 2a (derived) versus 2+ (ancestral) arrangement. Each cell is a count of sites (colour, $\log_{10}$). In the inversion (a) density spreads off the diagonal and into the corners, with variants common in one arrangement and absent in the other, i.e. near-fixed differences (3,828 sites, 18% shared polymorphism), the direct signature of the recombination barrier. In the collinear control (b) the density collapses onto the diagonal (0 near-fixed differences, 47% shared), as expected for freely recombining, exchangeable sequence.

Table S4: **Genes captured by the chromosome-2 inversion** (chr2:60.54–79.50 Mb). The 45 functionally annotated genes (of 89 gene models), ordered by position. The column fixed is the number of exonic fixed differences between the 2a and 2+ arrangements ($|\Delta AF| \geq 0.9$, '–' for none). The column sums to 24 of the 26 exonic fixed differences, and the other two fall in the unannotated gene model `00005332`. EnTAP best-hit descriptions. The set spans lipid metabolism (phospholipases, PNPLA, malonyl-/acyl-CoA transferases), peroxisomal and mitochondrial function, DNA repair (XRCC1, TDG, WRNIP1), transcription/splicing (ERG, PRPF8, WDR82) and signalling (PRKCD), with no coherent pathway, consistent with the absent GO enrichment (fig. S13). Gene IDs are the `LOC_` suffix.

| region (Mb) | gene | EnTAP best hit | fixed |
|---|---|---|---|
| 2:60.6 | `00005299` | protein HGH1 homolog isoform X1 | – |
| 2:60.9 | `00005300` | Phospholipase A2 domain-containing protein | 3 |
| 2:61.4 | `00005301` | lon protease homolog 2, peroxisomal isoform X2 | – |
| 2:61.4 | `00005302` | 39S ribosomal protein L18, mitochondrial | – |
| 2:61.4 | `00005303` | sodium- and chloride-dependent taurine transporter | – |
| 2:61.6 | `00005304` | cytochrome P450 4X1-like | 5 |
| 2:61.6 | `00005305` | uncharacterized protein LOC115219520 | – |
| 2:64.1 | `00005315` | PNPLA domain-containing protein | – |
| 2:64.1 | `00005316` | PNPLA domain-containing protein | – |
| 2:65.5 | `00005319` | CCHC-type domain-containing protein | – |
| 2:65.7 | `00005320` | uncharacterized protein LOC127831186 | – |
| 2:66.1 | `00005322` | solute carrier family 66 member 2 isoform X2 | – |
| 2:66.3 | `00005324` | Ubiquitin-conjugating enzyme E2 T | – |
| 2:68.0 | `00005330` | PCDHD1 (protocadherin) | – |

| region (Mb) | gene | EnTAP best hit | fixed |
|---|---|---|---|
| 2:68.8 | 00005334 | retrovirus-related Pol polyprotein (transposon) | – |
| 2:69.0 | 00005337 | PNPLA domain-containing protein | – |
| 2:69.0 | 00005338 | ELMO domain-containing protein | – |
| 2:69.1 | 00005339 | malonyl-CoA-acyl carrier protein transacylase, mito. | 1 |
| 2:69.1 | 00005340 | acidic repeat-containing protein-like | 6 |
| 2:69.1 | 00005341 | RPGRIP1L (ciliary) | 2 |
| 2:70.2 | 00005347 | transcriptional regulator ERG homolog | – |
| 2:71.2 | 00005349 | transcriptional regulator ERG homolog | – |
| 2:71.6 | 00005350 | uncharacterized protein LOC115215517 | 3 |
| 2:71.9 | 00005351 | cysteine-rich venom protein | 2 |
| 2:72.0 | 00005352 | cysteine-rich venom protein | – |
| 2:74.6 | 00005356 | DUF7043 domain-containing protein | – |
| 2:77.7 | 00005360 | SWIM-type domain-containing protein | – |
| 2:77.8 | 00005361 | UPB1 ($\beta$-ureidopropionase) | – |
| 2:77.8 | 00005362 | DNA-repair protein XRCC1 (N-terminal domain) | – |
| 2:77.8 | 00005363 | TIM21 (mitochondrial import) | – |
| 2:77.9 | 00005364 | dynein light chain roadblock-type 2 | – |
| 2:77.9 | 00005365 | TDG (thymine-DNA glycosylase) | – |
| 2:78.0 | 00005367 | small integral membrane protein 4 | – |
| 2:78.1 | 00005368 | uncharacterized protein LOC115226124 | – |
| 2:78.1 | 00005369 | ATPase WRNIP1-like | – |
| 2:78.2 | 00005371 | unnamed protein product (*Sepia*) | 1 |
| 2:78.2 | 00005372 | PPM1H (protein phosphatase) | – |
| 2:78.2 | 00005373 | WDR82 | – |
| 2:78.2 | 00005374 | glycerate kinase-like | – |
| 2:78.4 | 00005375 | PRKCD (protein kinase C $\delta$) | – |
| 2:78.6 | 00005376 | acyl-coenzyme A amino acid N-acyltransferase 1-like | – |
| 2:78.7 | 00005378 | uncharacterized protein LOC115226077 | – |
| 2:78.7 | 00005379 | PRPF8 (pre-mRNA splicing factor) | 1 |
| 2:78.8 | 00005380 | synaptic vesicle 2-related protein-like | – |
| 2:79.3 | 00005384 | unnamed protein product (*Sepia*) | – |

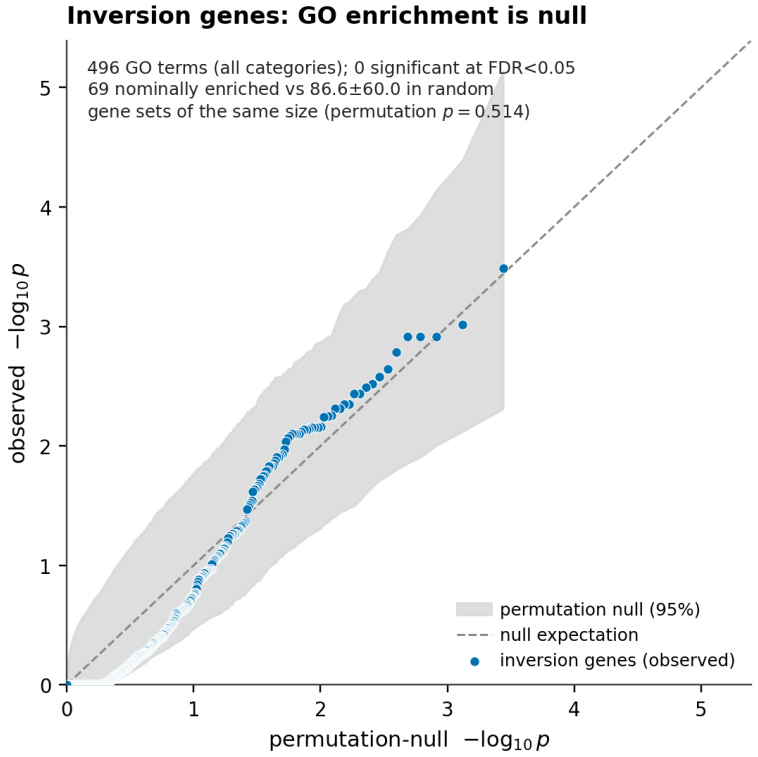



Figure S13: **The inversion-captured genes show no GO enrichment.** QQ-plot of the per-term GO-enrichment $p$-values (Fisher's exact vs. genome background, all three ontologies) for the genes in the inversion against a permutation null (500 random gene sets of the same size). A theoretical-uniform null is inappropriate for GO, because nested and correlated terms and tiny discrete counts inflate any small gene set, so the permutation band is the correct reference. The observed points fall entirely within it, and the inversion genes yield fewer nominally enriched terms than random sets of the same size (69 vs $86.6 \pm 60$, permutation $p = 0.51$) and none survive FDR<0.05. The large per-term fold values (up to $\sim$40$\times$) are small-count artefacts (most rest on 2 of the 29 GO-annotated genes), not enrichment (table S4).

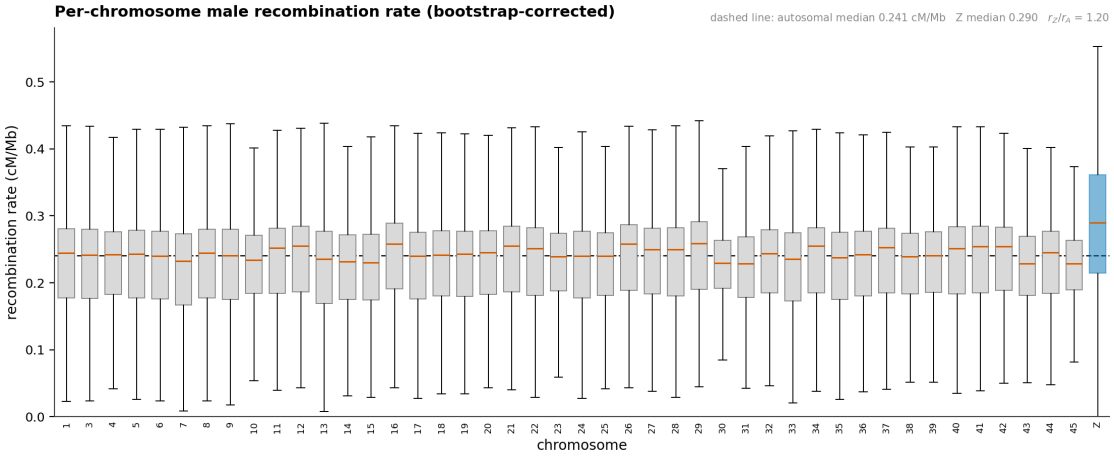



Figure S14: **Per-chromosome recombination-rate distribution** (ReLERNN, estimated in the 329 confidently sexed males). Autosomal rates from linkage disequilibrium are sex-averaged population rates, even when estimated in males. Rates are corrected with 200 parametric-bootstrap replicates in each of 20 rate slices. The autosomes were predicted in 3–6-kb windows by one network, trained on chromosomes 1, 13 and 45 and applied to every autosome except chromosome 2. The Z was predicted in 30-kb windows by a separate network trained with the Z effective size set to 0.60 of the autosomal value (the Z-to-autosome diversity ratio), which places its rate on the autosomal scale. The dashed line marks the median of all autosomal windows (0.241 cM/Mb), and per-chromosome medians span 0.229–0.259 cM/Mb. The Z median is 0.290 cM/Mb, so $r_Z/r_A = 1.20$. Because the Z recombines only in ZZ males, equal recombination in the two sexes predicts $r_Z/r_A = 0.67$ and achiasmate female meiosis predicts 1.33. Chromosome 2 is not shown, because a window-scale map estimated from all karyotypes together cannot resolve a barrier that acts only between arrangements. That barrier is shown directly by the linkage-disequilibrium block spanning the inversion (Supplementary Fig. S8).

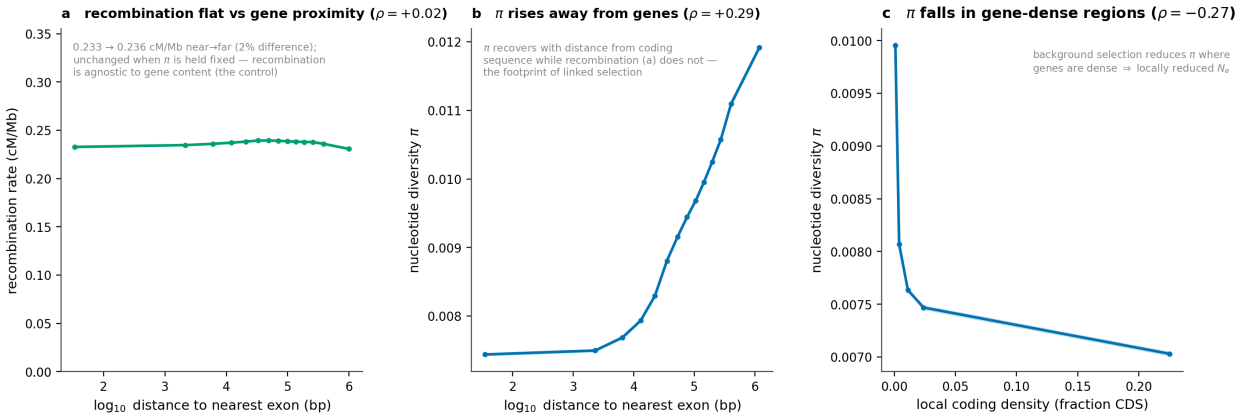



Figure S15: **Recombination, gene proximity and the efficacy of selection.** Per-window statistics (∼50 kb windows, autosomes). **(a)** As a control, recombination rate is essentially flat across distance from the nearest exon (0.233→0.236 cM/Mb near→far, a ∼2% difference, Spearman $\rho = +0.02$, unchanged when $\pi$ is held fixed, partial $\rho = -0.01$), so recombination is agnostic to gene content, as it should be. **(b)** Nucleotide diversity $\pi$, by contrast, rises steeply with distance from coding sequence ($\rho = +0.29$, +43% near→far), the footprint of linked selection. **(c)** $\pi$ falls with local coding density ($\rho = -0.27$), as background selection reduces diversity where genes are dense, i.e. a locally reduced effective size, and it is gene-density-driven. Linked selection in *Illex illecebrosus* is thus structured by gene density and gene proximity rather than by the recombination-rate axis, which is nearly uniform genome-wide (per-chromosome medians 0.23–0.26 cM/Mb, Supplementary Figs. S14, S16).

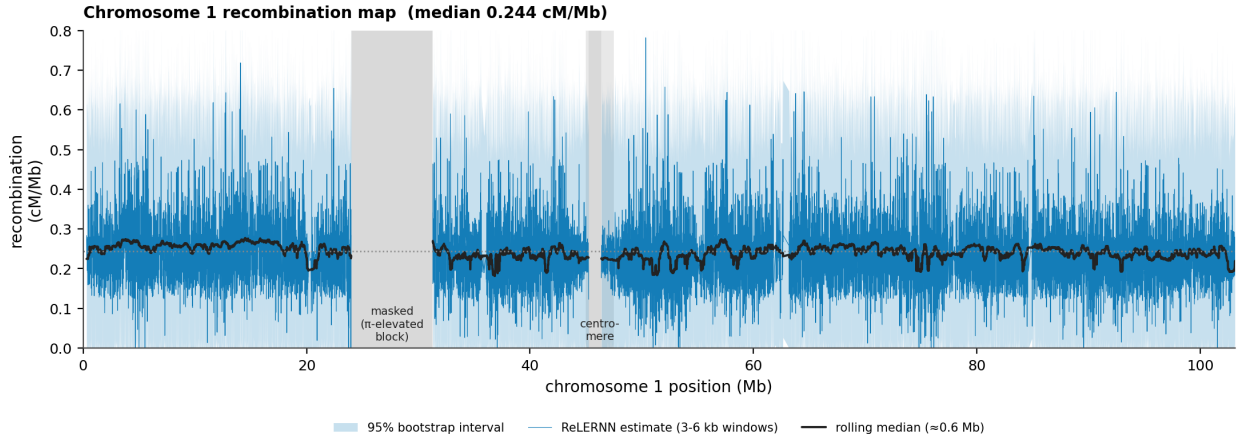



Figure S16: **The chromosome-1 recombination map, and what the LD-based map does not resolve.** Male ReLERNN estimate in 3–6-kb windows, corrected with 200 bootstrap replicates, with 95% bootstrap intervals and a 0.6-Mb rolling median (median 0.244 cM/Mb). Two stretches carry no windows. The $\pi$-elevated block at 24.0–31.3 Mb was masked. The 1.1-Mb gap at 45.25–46.39 Mb lies inside the 2.5-Mb block at 45.0–47.5 Mb, which is 80% repeat sequence (vs 52% chromosome-wide) and is the centromere. In the accessible pericentromeric flanks the estimate dips only ∼7%, although nucleotide diversity halves there (0.0038 vs 0.0084), and across the whole 103 Mb the 2-Mb rolling median spans just 0.22–0.27 cM/Mb (−10% to +10% of the median). The map's chromosome-scale dynamic range is therefore compressed, so centromeric suppression cannot be resolved and the nearly uniform landscape is in part an estimator ceiling.

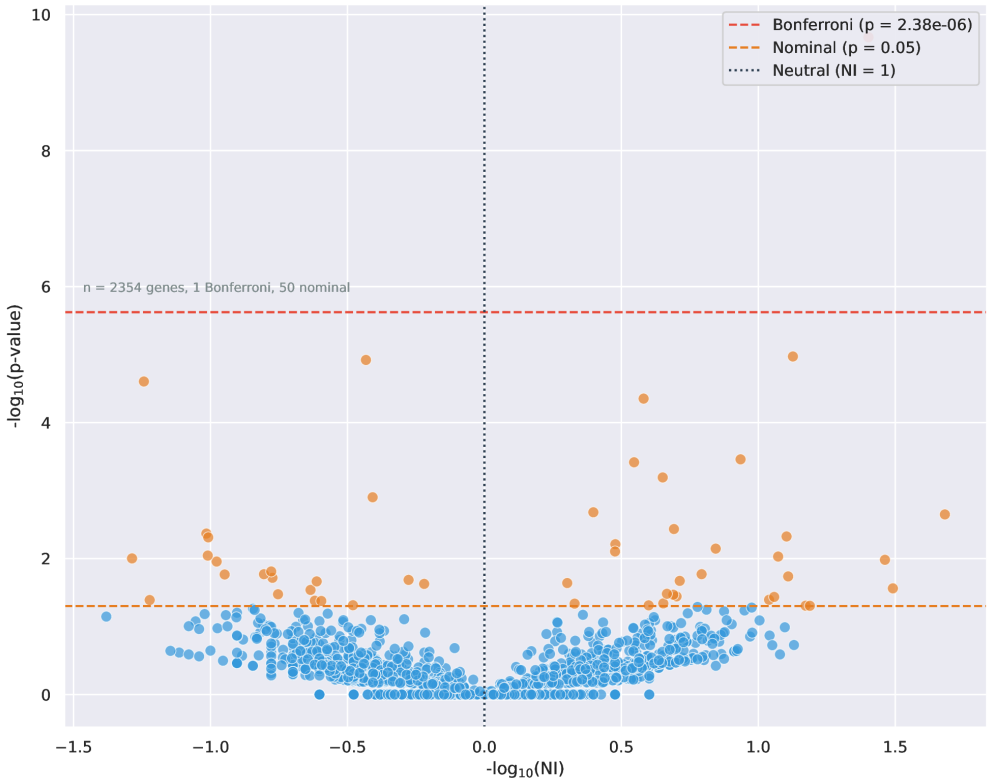



Figure S17: **Per-gene McDonald–Kreitman neutrality index** (mkado volcano plot, *I. coindetii* outgroup). Each point is one gene, plotted as $-\log_{10}$ of its neutrality index $\mathrm{NI} = (P_n/P_s)/(D_n/D_s)$ on the $x$-axis (right of the neutral $\mathrm{NI} = 1$ line = adaptive, $\mathrm{NI} < 1$, left = excess non-synonymous polymorphism, $\mathrm{NI} > 1$) against $-\log_{10} p$ on the $y$-axis. Dashed lines mark the nominal ($p = 0.05$) and Bonferroni thresholds. Of the 20,999 genes in the test, the 2,354 with non-zero $D_n$, $D_s$, $P_n$ and $P_s$ have a finite NI and are plotted. Of these, 51 are nominally significant, and one of them also passes Bonferroni correction over all 20,999 genes ($p < 2.4 \times 10^{-6}$). The genome-wide asymptotic $\alpha = 0.64$ resolves to few individual genes, as expected for a large-$N_e$ species where per-gene MK is underpowered.

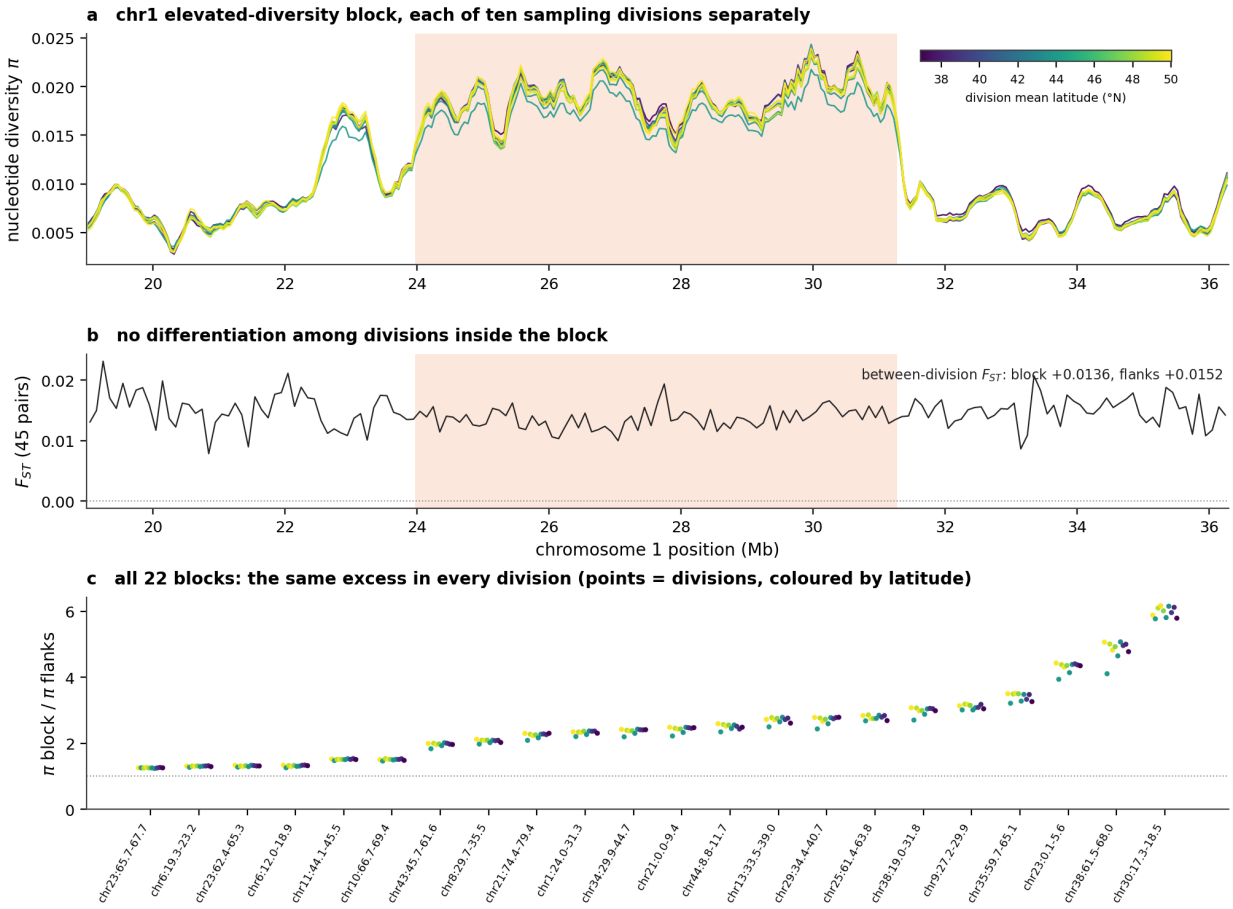



Figure S18: **The 22 elevated-diversity blocks are shared by every sampling division.** **(a)** Nucleotide diversity across chr1:19–36 Mb computed separately in each of the ten divisions with more than one individual (ANGSD, 50-kb windows, lines coloured by division latitude). The 24.0–31.3 Mb block (shaded) carries 2.2–2.4 times the flanking diversity in every division. **(b)** Between-division Hudson $F_{ST}$ over the same span (all 45 division pairs, 100-kb windows) is lower inside the block than in the flanks (0.014 vs 0.015). The same holds for the three strongest blocks (chr30:17.3–18.5, chr23:0.1–5.6 and chr38:61.5–68.0 Mb, 0.006–0.007 inside vs 0.014–0.015 outside). **(c)** Block-to-flank diversity ratio per division for all 22 blocks. The excess varies by only 1–6% among divisions, so no block is driven by a subset of the sample. Read depth separates the blocks into collapsed duplications (depth raised in every individual, up to 2.2 times the flanks in the population sample and 2.7 and 2.4 times in long reads from the assembly individual and a second individual, with five to six times the flanking density of heterozygous sites at skewed allele fractions in single diploid genomes), under-mapped blocks (short-read depth 0.6–0.9 times the flanks, a third of the accessible sequence not uniquely mappable, and a much smaller deficit in long reads), and gene-poor regions of elevated diversity with normal depth (Supplementary Table S5). Genotype-likelihood diversity at mapping quality 60 retains most of the excess (for example chr30 6.4 to 5.8 times, chr23 5.2 to 4.0 times), because reads from unassembled copies map uniquely to the single assembled copy. All 22 blocks are masked from the sweep scan and gene-set analyses.

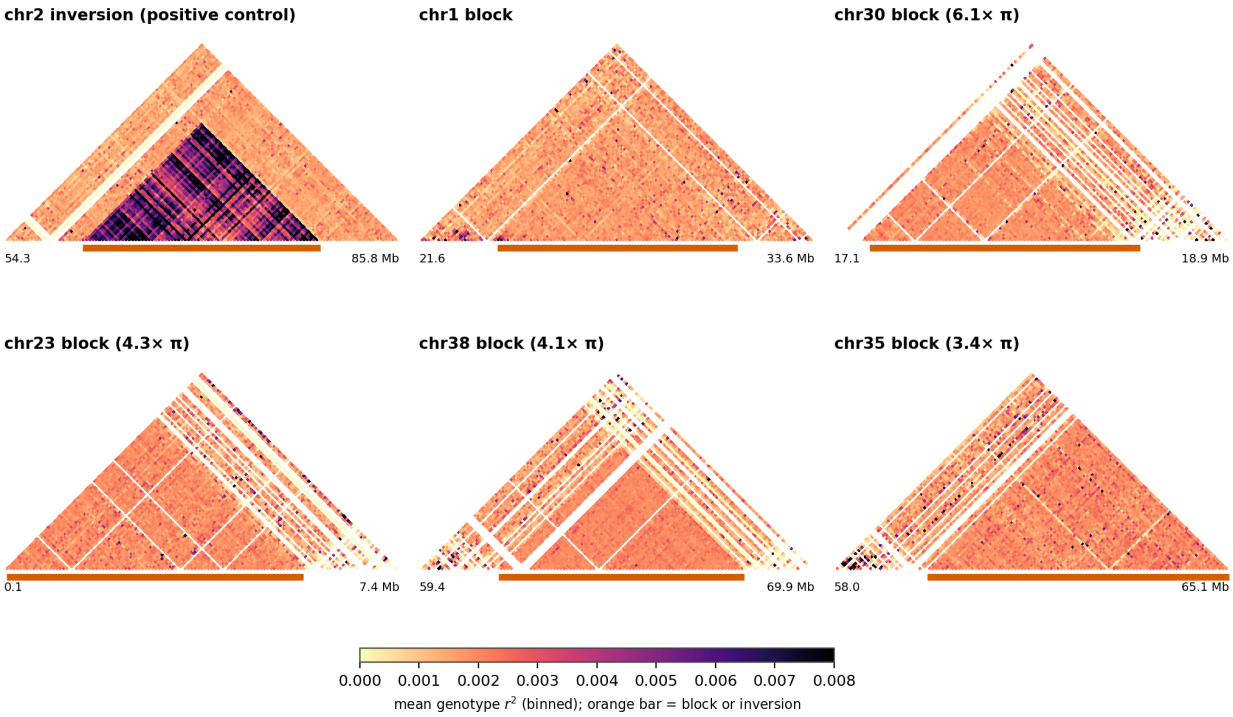



Figure S19: **The elevated-diversity blocks carry no inversion-like linkage disequilibrium.** Genotype $r^2$ among common SNPs (minor-allele frequency $\geq$0.1, $\leq$30% missing, 1,500 SNPs per region), averaged in 120 bins, for the chromosome-2 inversion (positive control) and five of the strongest blocks, each shown with a third of its length of flanking sequence. The orange bar marks the inversion or block. The inversion is a solid block of long-range LD over exactly its span (mean $r^2$ at >1 Mb 0.0056 inside vs 0.0016 in its flanks). None of the blocks differs from its flanks (0.0017–0.0020 vs 0.0012–0.0021 at >1 Mb, across all eight blocks tested), so none behaves as a segregating inversion or other long non-recombining haplotype. White stripes are bins with too few sampled SNPs.

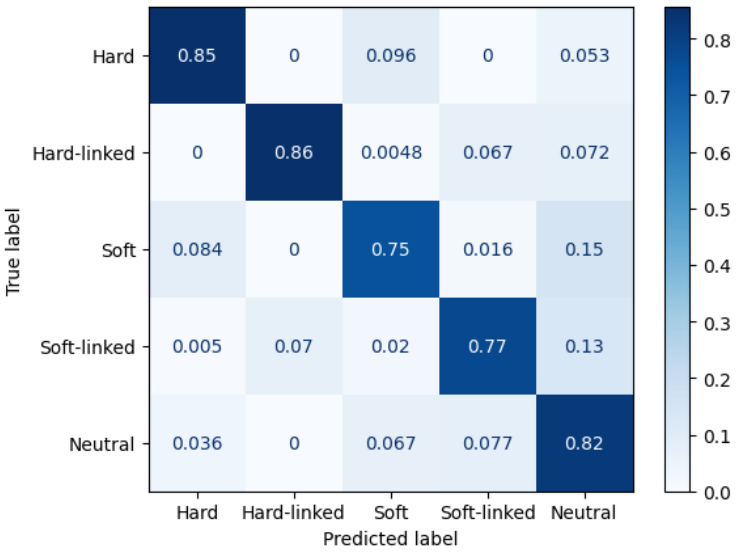



Figure S20: **Sweep-classifier confusion matrix** (five-class diploSHIC, full-SFS model, 81% overall accuracy). Recall is 0.85 for hard, 0.86 for linked-hard, 0.82 for neutral, 0.75 for soft and 0.77 for linked-soft sweeps.

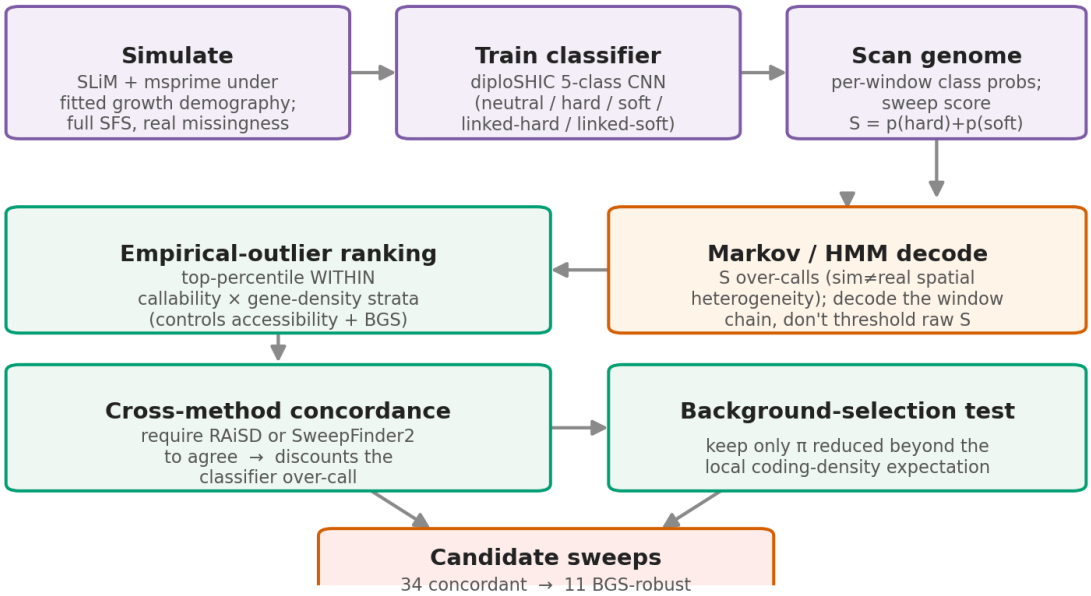



Figure S21: **The selective-sweep scan pipeline.** Simulate (SLiM + msprime under the fitted growth demography) → diploSHIC five-class classifier → Markov/HMM decode → stratified empirical-outlier ranking (within callability × gene-density strata) + cross-method concordance (RAiSD, SweepFinder2) + background-selection filter → candidate sweeps.

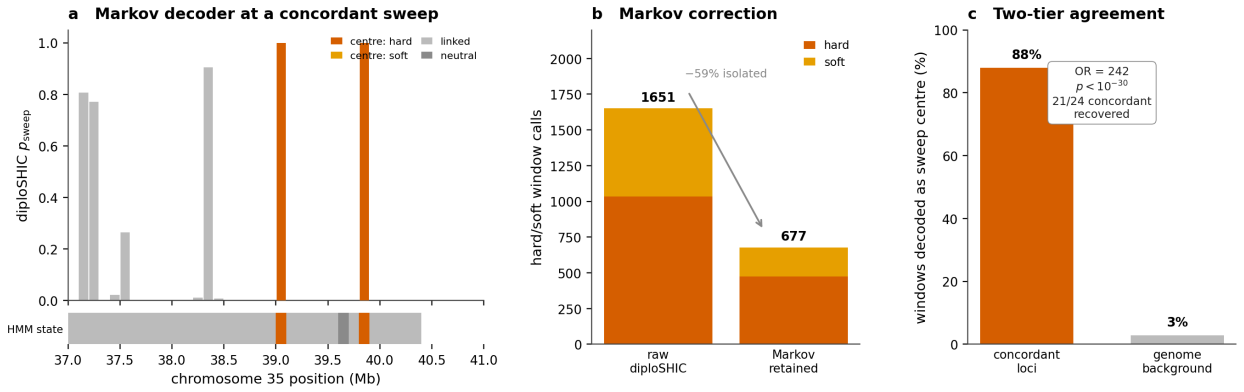



Figure S22: **Recombination-aware Markov decoder for the Tier-1 sweep calls.** A hidden Markov model over the per-100 kb-window diploSHIC classification (emissions = prior-corrected class posteriors, dwell lengths from the ∼400 kb sweep footprint scaled by the ReLERNN map, design-based rather than simulation-calibrated). **(a)** The decoder at a representative concordant sweep (chromosome 35, *ZEB2*), showing the sweep posterior $p_{\text{hard}} + p_{\text{soft}}$ per window and the Viterbi state track (neutral / linked / centre). It localises the centre inside the linked envelope and overrides spatially isolated high-posterior windows (e.g. near 38.35 Mb). **(b)** Markov correction reduces the raw hard/soft windows from 1,651 to 677 spatially coherent windows (−59%), which merge into 623 localized autosomal calls (the four Z calls come from the separate males-only scan). **(c)** 88% of windows at the Tier-2 cross-method-concordant loci are decoded as sweep centres versus 2.9% genome-wide (odds ratio 242, $p < 10^{-30}$, 21 of 24 concordant regions recovered).

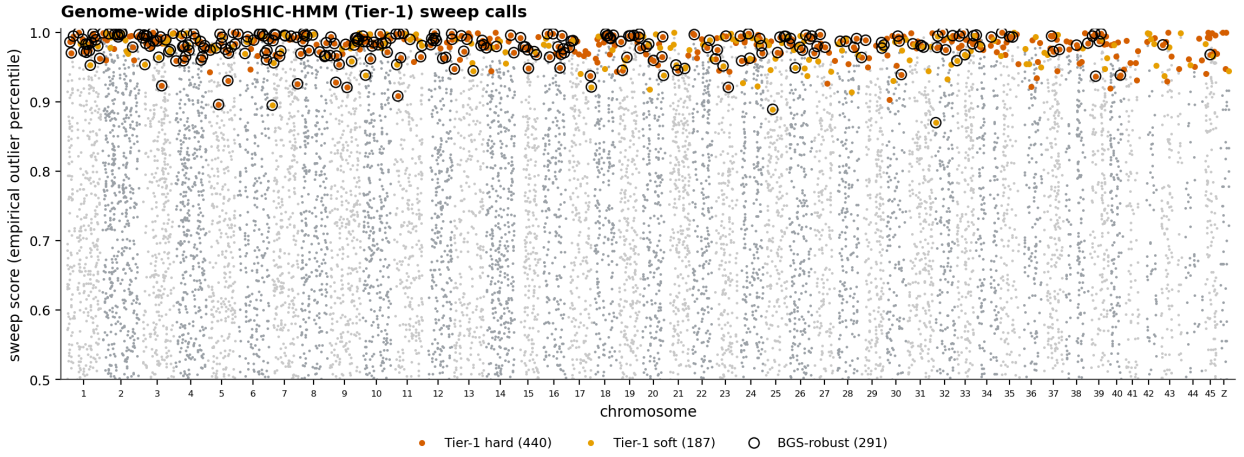



Figure S23: **Genome-wide diploSHIC-HMM (Tier-1) sweep calls.** Per-100 kb-window empirical-outlier sweep score (rank within callability × coding-density strata) across all 45 autosomes plus the Z chromosome (627 calls in total, 440 hard and 187 soft). The Markov-decoded Tier-1 calls are highlighted by class (hard, soft), and the background-selection-robust subset (§ BGS screen, 291 of the 623 autosomal calls) is ringed. The four chromosome-Z calls come from the male-only ($n$=329) diploSHIC scan and are diploSHIC-only (RAiSD/SweepFinder2 do not cover the Z). The calls are distributed across the genome with no single dominant locus.

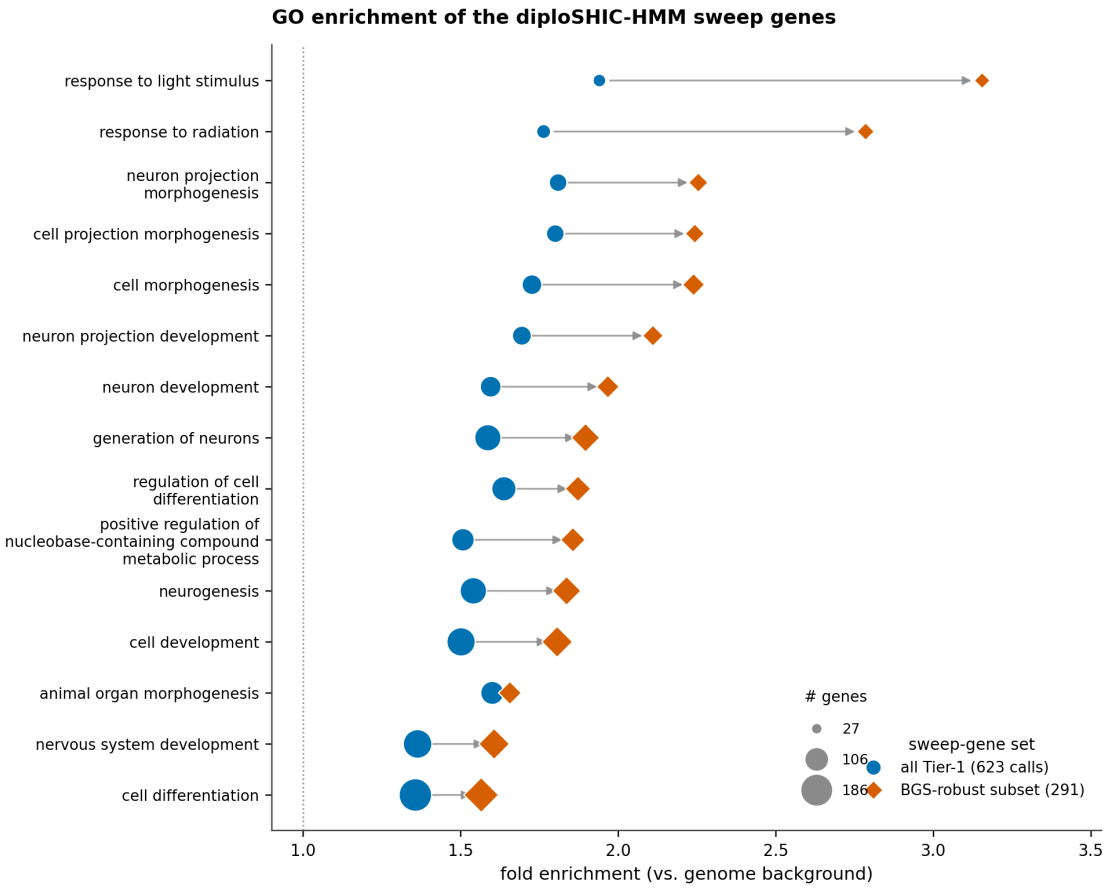



Figure S24: **GO enrichment of the diploSHIC-HMM sweep genes** (biological process, terms significant at FDR<0.05 in both sets). In this paired dot plot each term is shown for the full Tier-1 set (blue circle, 623 calls) and for its background-selection-robust subset (orange diamond, 291 calls), with $x$ = fold enrichment and marker area = number of sweep genes carrying the term. The grey arrow marks the shift between the two sets. The BGS screen consistently sharpens the enrichment (rightward), and most strongly for the light-response / photoreception terms (response to light stimulus 1.9 $\rightarrow$ 3.2×, response to radiation 1.8 $\rightarrow$ 2.8×). Two coherent themes dominate, neural development and photoreception, and both survive the background-selection filter (table S6, full per-set enrichment tables are provided with the source). The hard- and soft-sweep subsets are tested separately in table S6.

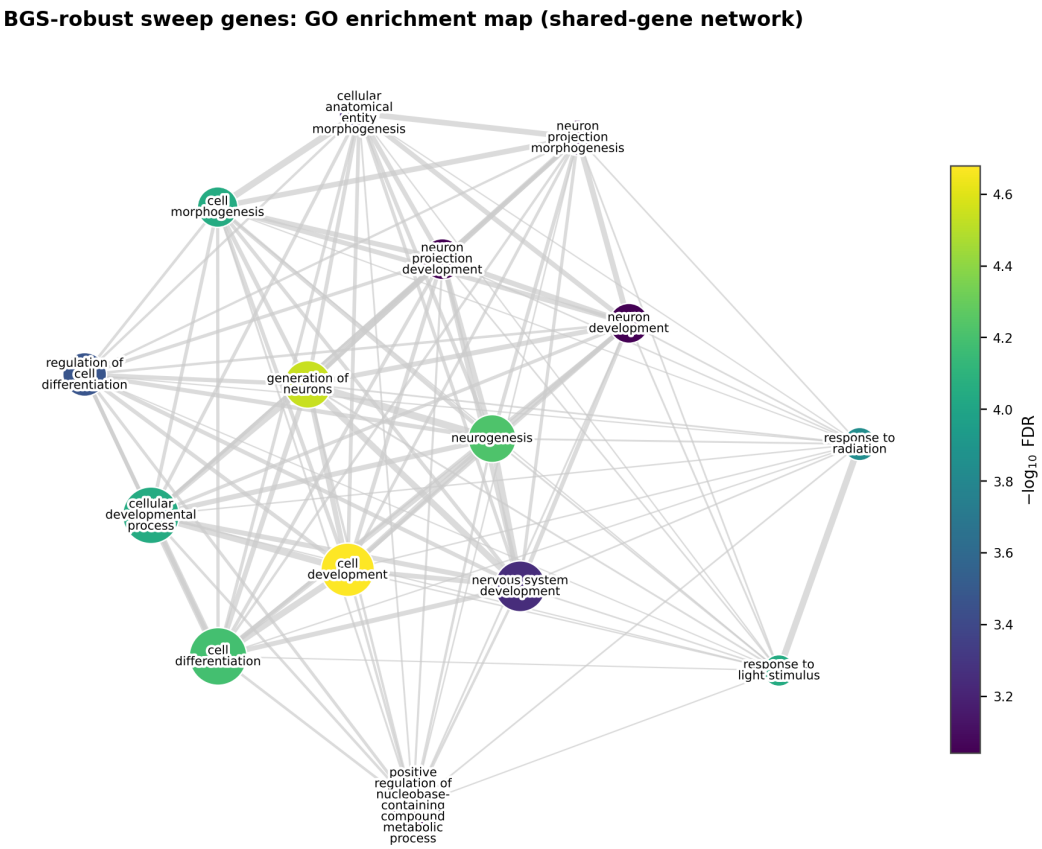



Figure S25: **GO enrichment map for the background-selection-robust sweep genes.** Built from the 291 BGS-robust Tier-1 calls (the subset whose local diversity reduction exceeds the background-selection expectation, 227 GO-annotated genes). Nodes are enriched biological-process terms (size = number of genes, colour = $-\log_{10}$ FDR). Edges connect terms sharing genes (Jaccard $\geq 0.15$). The terms form connected modules of neural development / cell morphogenesis and photoreception (response to light/radiation), the same two themes seen in the GSEA dot plot (fig. S24).

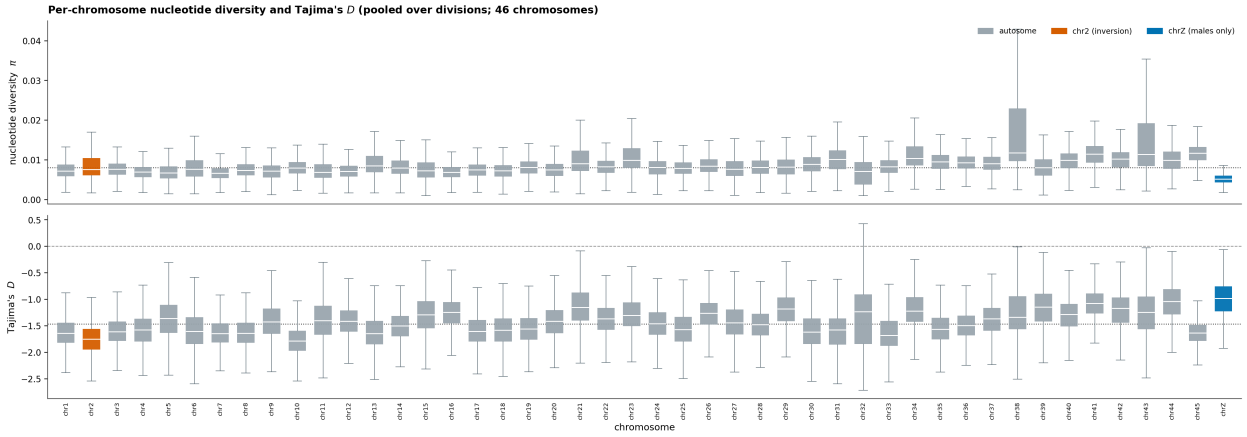



Figure S26: **Per-chromosome diversity.** Distribution of nucleotide diversity $\pi$ (top) and Tajima's $D$ (bottom) across 50 kb windows, one box per chromosome (pooled over the ten NAFO divisions, for all 46 chromosomes, the 45 autosomes plus chrZ). Diversity is uniform and $D$ is strongly negative genome-wide, the signature of expansion. Two chromosomes stand apart. Chromosome 2 (the inversion, orange) carries an inflated upper tail of $\pi$ and more-negative $D$ from the segregating arrangements, and chrZ (blue, computed in the 329 hemizygous-informative males only, as females are Z-haploid) has reduced diversity ($\pi$ median 0.005 vs $\sim$0.008 autosomal, $\approx$0.63$\times$) as expected for a sex chromosome.

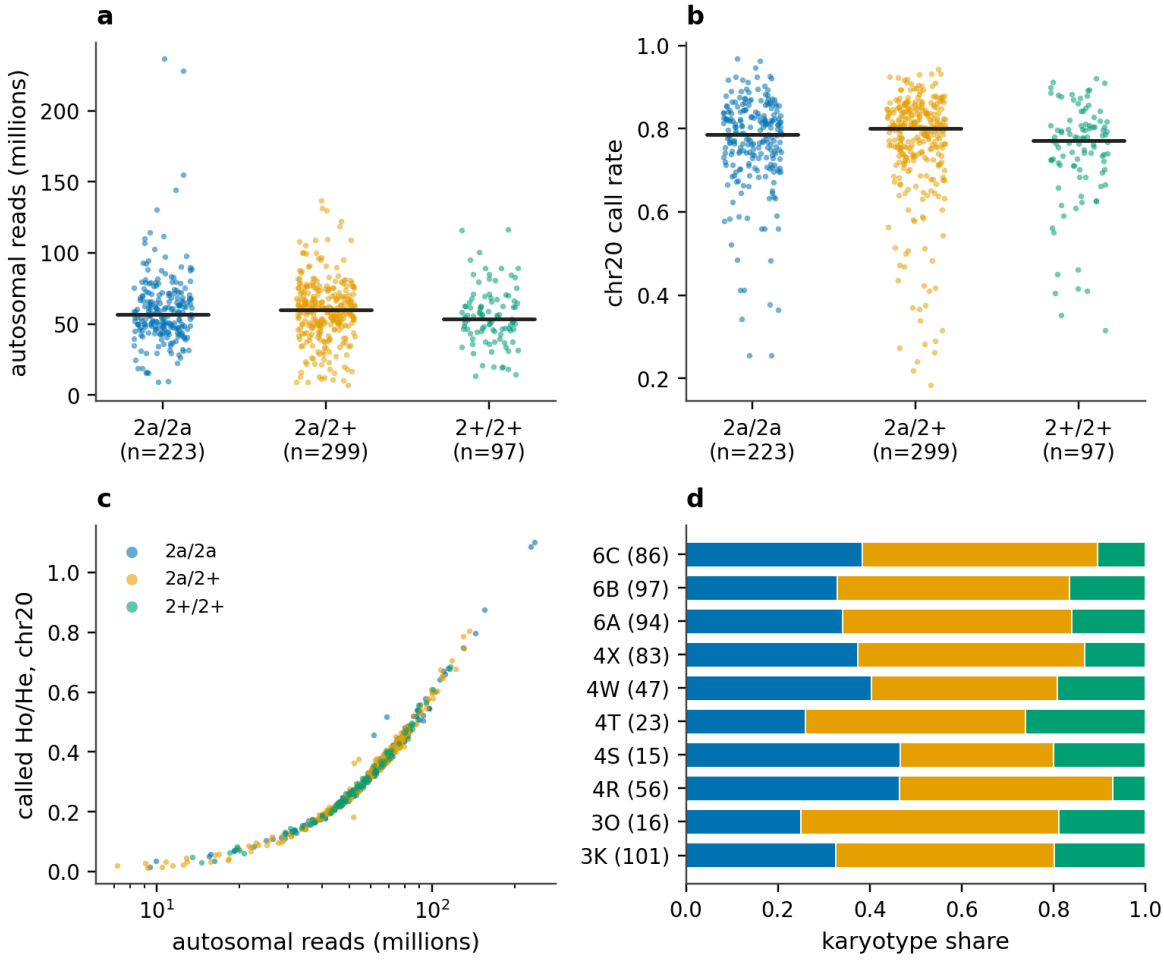



Figure S27: **Sequencing depth, call rate and sampling by chromosome-2 karyotype. (a)** Autosomal reads per individual (samtools idxstats) and **(b)** call rate at the biallelic chromosome-20 SNPs with minor-allele frequency $\geq$0.05, by karyotype group, with medians. **(c)** Called $H_o/H_e$ on chromosome 20 against read count. Heterozygotes are under-called at low depth in every group alike. **(d)** Karyotype composition of the ten divisions with at least five individuals (numbers in parentheses).

Table S3: **Use of the outgroups.** *I. coindetii* served three roles, the structural comparison of the chromosome-2 arrangements, a count of allele sharing at sites that differ between them, and the divergence term of the genome-wide McDonald–Kreitman (MK) test, the latter resting on a whole-genome *I. coindetii*↔*Illex illecebrosus* AnchorWave alignment. *I. argentinus* was aligned identically (CDS) and used as a secondary MK outgroup, and its reads were counted at the sites that differ between the arrangements. It yields ∼9× fewer fixed coding differences than *I. coindetii* yet has genome-wide $\pi$ indistinguishable from *Illex illecebrosus*, marking it as a recently-diverged congener of near-identical diversity. *S. oualaniensis* was used only for the structural comparison at the breakpoints, with positions given on the primary assembly and on HiC_scaffold_1.

| Step | Quantity | Value |
|---|---|---|
| Alignment (AnchorWave v1.2.3, CDS-anchored) | | |
| | *I. coindetii* substitution sites (genome-wide) | 45,722,335 |
| | aligned scaffold identity (inversion) | 99.2–99.9% |
| Chromosome-2 arrangements in the reference individual (minimap2) | | |
| | hap1 vs. primary, inverted segment | 60.54–79.53 Mb |
| | hap2 vs. primary | collinear |
| | arrangement of hap1 / hap2 | 2a / 2+ |
| | *I. coindetii* 37.95–74.95 Mb, aligned to 2+ | one inverted run |
| | … aligned to 2a | two runs, split at 60.5 Mb |
| | *I. coindetii* adjacency at 56.9 Mb | kept in 2+, broken in 2a |
| Chromosome-2 breakpoints in *S. oualaniensis* (HiC_scaffold_1, minimap2) | | |
| | *Illex illecebrosus* chr2 genes transferred to HiC_scaffold_1 | 265 of 319 |
| | left breakpoint, genes at 60.49 and 60.55 Mb | adjacent (51.52, 51.57 Mb) |
| | right breakpoint, 78.2–83.6 Mb | one inverted run |
| | 2+ adjacencies (hap2), both breakpoints | present |
| | 2a adjacencies (hap1), both breakpoints | absent |
| Allele sharing with *I. coindetii* (inversion, frequency difference between 2a and 2+ $\geq 0.7$) | | |
| | differentiated sites (*I. coindetii* base callable) | 15,093 (14,650) |
| | … 2a majority allele = *I. coindetii* | 26.0% |
| | *I. coindetii* differs from the reference base | 4,582 |
| | … 2a / 2+ carries the *I. coindetii* allele | 65.0% / 35.0% |
| *I. argentinus* reads at the same sites (10 individuals, ∼0.3× each) | | |
| | sites covered / reads | 13,195 / 34,860 |
| | reference bias, collinear control | $0.018 + 1.02p$ |
| | 2a frequency, pooled (per individual) | 0.284 (0.27–0.30) |
| | sites with ≥4 reads showing one allele | 87% |
| | 2a allele where *I. coindetii* carries 2a / 2+ | 0.68 / 0.24 |
| McDonald–Kreitman test (mkado, *I. coindetii* divergence) | | |
| | genes tested | 20,999 |
| | nonsynonymous / synonymous divergence (Dn / Ds) | 15,343 / 13,670 |
| | nonsynonymous / synonymous polymorphism (Pn / Ps) | 468,124 / 630,949 |
| | standard $\alpha$, pooled counts | 0.339 |
| | asymptotic $\alpha$ (95% bootstrap CI) | 0.642 (0.630–0.656) |
| I. argentinus (secondary MK outgroup, CDS alignment) | | |
| | nonsynonymous / synonymous divergence (Dn / Ds) | 1,355 / 1,805 |
| | fixed coding differences (vs *I. coindetii* 29,013) | 3,160 (∼9× fewer) |
| | asymptotic $\alpha$ | 0.526 |
| | genome-wide $\pi$ (ANGSD, 6 autosomes; *Illex illecebrosus* = 0.0093) | 0.0092 |

Table S5: **Anomalous-diversity blocks, masked from the sweep scan and gene-set analyses.** Blocks of ≥1 Mb whose rolling nucleotide diversity exceeds twice the chromosome median. $\pi$ fold is the block mean over the chromosome median. Division ratio is the range of block-to-flank diversity across the ten sampling divisions (ANGSD). Depth is block-to-flank read depth, as the median over the 621 individuals with data (Illumina, mapping quality ≥0) and for the assembly individual (PacBio HiFi, mapping quality ≥20). Non-unique is the percentage of the block's accessible sequence that 150-bp reads cannot map uniquely (GenMap, up to two mismatches), against 3.8% genome-wide. Class is collapsed duplication (population depth ≥1.15), under-mapped (depth ≤0.90), or elevated diversity with normal depth. In long-read genotypes of two individuals, collapsed blocks carry five to six times the flanking density of heterozygous sites at skewed allele fractions (median minor-allele fraction 0.29–0.33 vs 0.38–0.42 in flanks), and elevated-diversity blocks carry about twice the flanking density at normal fractions (chr1, 0.41 vs 0.40–0.43). The under-mapped chr21:0.0–9.4 block also shows skewed fractions (0.25 and 0.28 vs 0.42 and 0.44), with heterozygous-site density halved in the assembly individual, whose second haplotype largely fails to map, and tripled in the second individual, consistent with under-mapped blocks being largely duplicated within the assembly (33% of their accessible sequence non-unique).

| Block | Mb | $\pi$ fold | $D$ block / chrom | Genes/Mb | Division ratio | Depth pop. / HiFi | Non-unique (%) | Class |
|---|---|---|---|---|---|---|---|---|
| chr1:24.0–31.3 | 7.3 | 2.8 | −1.77 / −2.21 | 1.8 | 2.20–2.37 | 0.97 / 1.05 | 7 | elevated diversity |
| chr6:12.0–18.9 | 6.9 | 2.3 | −1.58 / −2.17 | 1.7 | 1.26–1.34 | 0.62 / 0.92 | 45 | under-mapped |
| chr6:19.3–23.2 | 3.9 | 2.4 | −1.56 / −2.17 | 1.8 | 1.27–1.32 | 0.66 / 0.84 | 41 | under-mapped |
| chr8:29.7–35.5 | 5.8 | 2.3 | −1.97 / −2.22 | 1.7 | 1.97–2.12 | 0.98 / 1.10 | 7 | elevated diversity |
| chr9:27.2–29.9 | 2.7 | 3.6 | −1.12 / −2.12 | 2.2 | 3.01–3.19 | 1.19 / 1.18 | 9 | collapsed duplication |
| chr10:66.7–69.4 | 2.7 | 2.1 | −2.06 / −2.26 | 0.4 | 1.46–1.53 | 0.95 / 1.01 | 2 | elevated diversity |
| chr11:44.1–45.5 | 1.4 | 2.1 | −1.83 / −2.15 | 2.1 | 1.48–1.53 | 0.97 / 1.04 | 3 | elevated diversity |
| chr13:33.5–39.0 | 5.5 | 2.7 | −1.79 / −2.19 | 3.1 | 2.50–2.79 | 0.93 / 1.05 | 4 | elevated diversity |
| chr21:0.0–9.4 | 9.4 | 2.6 | −1.39 / −1.94 | 1.7 | 2.22–2.49 | 0.68 / 0.61 | 36 | under-mapped |
| chr21:74.4–79.4 | 5.0 | 2.5 | −1.42 / −1.94 | 2.0 | 2.09–2.30 | 0.66 / 0.61 | 39 | under-mapped |
| chr23:0.1–5.6 | 5.5 | 4.3 | −0.96 / −2.04 | 2.2 | 3.94–4.43 | 1.78 / 1.91 | 12 | collapsed duplication |
| chr23:62.4–65.3 | 2.9 | 2.3 | −1.68 / −2.04 | 2.8 | 1.28–1.33 | 0.88 / 1.11 | 8 | under-mapped |
| chr23:65.7–67.7 | 2.0 | 2.3 | −1.73 / −2.04 | 1.5 | 1.23–1.27 | 1.07 / 1.12 | 10 | elevated diversity |
| chr25:61.4–63.8 | 2.4 | 2.8 | −1.57 / −2.17 | 2.9 | 2.68–2.86 | 1.16 / 1.16 | 25 | collapsed duplication |
| chr29:34.4–40.7 | 6.3 | 2.6 | −1.41 / −1.97 | 2.1 | 2.44–2.79 | 0.86 / 0.98 | 30 | under-mapped |
| chr30:17.3–18.5 | 1.2 | 6.1 | −0.69 / −2.17 | 2.5 | 5.77–6.17 | 2.19 / 2.66 | 2 | collapsed duplication |
| chr34:29.9–44.7 | 14.8 | 2.6 | −1.45 / −1.95 | 1.8 | 2.20–2.43 | 0.81 / 0.93 | 33 | under-mapped |
| chr35:59.7–65.1 | 5.4 | 3.4 | −1.47 / −2.13 | 1.9 | 3.21–3.52 | 1.53 / 1.84 | 4 | collapsed duplication |
| chr38:19.0–31.8 | 12.8 | 2.8 | −1.41 / −1.97 | 2.2 | 2.70–3.08 | 0.87 / 1.17 | 30 | under-mapped |
| chr38:61.5–68.0 | 6.5 | 4.1 | −1.08 / −1.97 | 2.6 | 4.11–5.08 | 1.44 / 1.21 | 17 | collapsed duplication |
| chr43:45.7–61.6 | 15.9 | 2.4 | −1.52 / −1.84 | 2.2 | 1.83–2.02 | 0.93 / 0.94 | 31 | elevated diversity |
| chr44:8.8–11.7 | 2.9 | 2.7 | −1.27 / −1.85 | 1.4 | 2.35–2.59 | 0.96 / 0.88 | 5 | elevated diversity |

Table S6: **Gene-ontology enrichment of the Tier-1 (diploSHIC-only, Markov-corrected) sweep genes.** The 623 Tier-1 calls overlap 885 genes (475 GO-annotated). Enrichment by Fisher's exact test against the annotated-gene background on the scanned chromosomes, Benjamini–Hochberg FDR. 124 terms (116 biological process, 6 molecular function and 2 cellular component) reach FDR<0.05, and representative terms are shown, spanning the two dominant themes (neural development and RNA-regulatory or transcriptional control). Every term shown also reaches FDR<0.05 in the hard-sweep gene set alone, which gives 98 terms, 85 of them shared with the full set. The soft-sweep gene set, with 142 GO-annotated genes against 333 in the hard-sweep set, has no term at FDR<0.05. The full machine-readable table is `hmm_decode_masked/tier1_GO_enrichment.tsv`.

| GO id | term (biological process) | genes (T1/bg) | fold | FDR |
|---|---|---|---|---|
| GO:0048699 | generation of neurons | 123/1441 | 1.59 | $2.0 \times 10^{-4}$ |
| GO:0022008 | neurogenesis | 125/1508 | 1.54 | $2.6 \times 10^{-4}$ |
| GO:0030182 | neuron differentiation | 91/1075 | 1.57 | $1.6 \times 10^{-3}$ |
| GO:0030154 | cell differentiation | 186/2548 | 1.36 | $5.4 \times 10^{-4}$ |
| GO:0009887 | animal organ morphogenesis | 96/1115 | 1.60 | $8.9 \times 10^{-4}$ |
| GO:0050767 | regulation of neurogenesis | 66/693 | 1.77 | $1.3 \times 10^{-3}$ |
| GO:0000902 | cell morphogenesis | 72/775 | 1.73 | $1.3 \times 10^{-3}$ |
| GO:0007409 | axonogenesis | 45/463 | 1.81 | $1.0 \times 10^{-2}$ |
| GO:0048858 | cell projection morphogenesis | 59/609 | 1.80 | $1.6 \times 10^{-3}$ |
| GO:0051252 | regulation of RNA metabolic process | 145/1920 | 1.40 | $1.3 \times 10^{-3}$ |
| GO:2001141 | regulation of RNA biosynthetic process | 135/1768 | 1.42 | $1.6 \times 10^{-3}$ |

Table S7: **Functional clusters of the background-selection-robust Tier-1 sweep genes.** The 151 GO terms enriched at FDR<0.05 in the BGS-robust set (137 biological process, 12 molecular function and 2 cellular component, fig. S24), each assigned to one functional cluster by keyword rules on the term name (`scripts/tableS5_go_clusters.py`). Transcriptional regulation and neural development are the largest clusters, followed by organ development and cell morphogenesis/differentiation, with a distinct photoreception/sensory cluster. Fold range and the representative term (lowest FDR) are given per cluster.

| functional cluster | terms | fold range | representative term (min FDR) |
|---|---|---|---|
| RNA / transcription regulation | 36 | 1.4–2.7 | pos. reg. nucleobase-containing compound metabolic process ($9 \times 10^{-4}$) |
| Neural development | 28 | 1.6–22.2 | generation of neurons ($3 \times 10^{-5}$) |
| Organ / anatomical development | 21 | 1.2–2.5 | anatomical structure morphogenesis ($2 \times 10^{-3}$) |
| Cell morphogenesis & differentiation | 13 | 1.5–2.2 | cell development ($2 \times 10^{-5}$) |
| Photoreception / sensory response | 7 | 1.5–13.8 | response to light stimulus ($9 \times 10^{-5}$) |
| Other regulation / metabolism | 46 | 1.2–11.4 | reg. response to stimulus ($3 \times 10^{-3}$) |

Table S8: **Tier-2 (cross-method-concordant) sweep genes.** The 24 genome-wide-scan regions where the diploSHIC-HMM sweep signal is independently corroborated by RAiSD, which combines the site-frequency spectrum, linkage disequilibrium and diversity, or by SweepFinder2, a site-frequency-spectrum likelihood method. Ten regions carry a named protein-coding gene (13 genes, listed). The remaining concordant regions are gene deserts or uncharacterised loci. At the two regions marked † the gene is the target assigned by fine-scale localisation of the diversity minimum, as in main-text Table 4. Coordinates are in Mb, and $\alpha$ and DoS are from the McDonald–Kreitman test against *I. coindetii* ('–' where the test is unpowered, i.e. too few fixed/polymorphic sites). The full candidate set, comprising all 227 diploSHIC-outlier regions, the 24 concordant regions and the per-region background-selection test, is provided as machine-readable tables (`analysis/steps/14_sweep_seqmodel/results/empirical_scan_fullsfs/outlier_scan_45_masked/{windows,regions}.tsv` and the `concordant` gene/MK/BGS tables).

| region (Mb) | sweep | gene | EnTAP best hit | $\alpha$ | DoS |
|---|---|---|---|---|---|
| 1:20.3 | hard | *Aplnr* | Apelin receptor | 1.00 | +0.17 |
| 1:85.5 | hard | *Cpes* † | Ceramide phosphoethanolamine synthase | – | -0.52 |
| 2:84.8 | hard | *NDRG3* | Protein NDRG3 | – | -0.18 |
| 5:38.0 | hard | *cept1* | Choline/ethanolaminephosphotransferase 1 | – | -1.00 |
| 5:38.0 | hard | *Arsi* | Arylsulfatase I | – | -0.25 |
| 6:3.0 | hard | *Rexo1* | RNA exonuclease 1 homolog | – | -0.53 |
| 13:73.2 | hard | *Rab3* | Ras-related protein Rab-3 | – | -0.17 |
| 13:73.2 | hard | *hyi* | Putative hydroxypyruvate isomerase | – | -0.64 |
| 13:73.2 | hard | *Vps28* | Vacuolar protein sorting-associated protein 28 | – | – |
| 25:65.0 | soft | *GALT* † | Galactose-1-phosphate uridylyltransferase | 0.07 | +0.02 |
| 32:9.3 | soft | *MACROD2* | ADP-ribose glycohydrolase MACROD2 | – | -0.59 |
| 33:9.0 | hard | *Tinagl1* | Tubulointerstitial nephritis antigen-like | – | -0.23 |
| 35:39.0 | hard | *ZEB2* | Zinc finger E-box-binding homeobox (ZEB) | – | -0.31 |

Table S9: **Z recombination and sex-biased mutation.** Top, the ratio of population-scaled recombination rates on the Z and the autosomes in the 329 ZZ males from the decay of $\sigma_D^2$ with distance (86 autosomal and nine Z windows of 2 Mb, 95% CI from 200 bootstrap resamples of windows). Bottom, the ratio $R$ of Z to autosomal synonymous divergence between *Illex illecebrosus* and *S. oualaniensis* and the male-to-female mutation ratio $\alpha = (3R-2)/(4-3R)$, following Miyata et al. (1987), for all genes and with autosomal genes matched to the Z genes by $d_N/d_S$, by third-position GC content (GC3) or by both, or restricted to genes above the autosomal median $d_N/d_S$ (95% CI from 2,000 bootstrap resamples of genes within strata). The $\alpha$ limits are the $R$ limits transformed, and an $R$ limit below 2/3 gives $\alpha < 0$, outside its range, so it is reported as 0.

| Linkage disequilibrium | $\rho_Z/\rho_A$ | 95% CI |
|---|---|---|
| Ohta–Kimura decay fit | 1.04 | 0.88–1.28 |
| Curve rescaling | 1.08 | 0.78–1.67 |

| Synonymous divergence | Z / autosomal genes | $R$ (95% CI) | $\alpha$ (95% CI) |
|---|---|---|---|
| All genes | 109 / 8,649 | 0.83 (0.72–0.94) | 0.33 (0.10–0.70) |
| Matched on $d_N/d_S$ | 109 / 8,649 | 0.84 (0.73–0.96) | 0.36 (0.11–0.79) |
| Above median $d_N/d_S$ | 44 / 4,324 | 0.79 (0.66–0.93) | 0.24 (0–0.65) |
| Matched on GC3 | 109 / 8,649 | 0.87 (0.76–0.98) | 0.43 (0.17–0.89) |
| Matched on $d_N/d_S$ and GC3 | 109 / 6,042 | 0.89 (0.78–1.01) | 0.49 (0.20–1.06) |

Table S10: **Software and versions.** Versions were read off the analysis machine (conda metadata, `pip list`, `--version`, or recorded `versions.txt`).

| Tool (version) | Used for | Reference |
|---|---|---|
| msinv 0.4.1 | chr2 inversion simulation (ABC-RF, NPE) | (Small, 2025; Kelleher et al., 2018) |
| sbi 0.27 | neural posterior estimation | (Tejero-Cantero et al., 2020; Boelts et al., 2025) |
| abcrf 2.0 (R) | ABC random-forest model choice | (Pudlo et al., 2016) |
| grenepipe 0.15.0 | read processing and variant calling | (Czech and Exposito-Alonso, 2022) |
| fastp (wrapper 0.64.0) | read trimming | (Chen et al., 2018) |
| BWA-MEM2 2.2.1 | read mapping | (Vasimuddin et al., 2019) |
| Picard MarkDuplicates 3.3.0 | duplicate removal | Broad Institute |
| GATK 4.6.1.0 | HaplotypeCaller and joint genotyping (gVCF, all sites) | (Poplin et al., 2017) |
| GenMap 1.3.0 | mappability (150-mers, 2 mismatches) | (Pockrandt et al., 2020) |
| plink2 2.00a6LM | per-division allele counts (per-SNP Hudson $F_{ST}$), KING-robust kinship, heterozygosity | (Chang et al., 2015; Manichaikul et al., 2010) |
| snpEff 5.4c | variant annotation | (Cingolani et al., 2012) |
| mosdepth 0.3.12 | GC-normalised read depth (high-depth mask) | (Pedersen and Quinlan, 2018) |
| LiftOn | annotation transfer to *S. oualaniensis* (version not recorded) | (Chao et al., 2025) |
| MACSE 2.07 | codon alignment (sex-biased mutation) | (Ranwez et al., 2018) |
| EnTAP | functional annotation (`Illex_F24` annotation, version not recorded) | (Hart et al., 2020) |
| ANGSD 0.940 | SFS, $\theta$, diversity (demography, diversity) | (Korneliussen et al., 2014) |
| Stairway Plot 2 (2.1.1) | demography | (Liu and Fu, 2020) |
| GONE2 1.0.4 | recent demography | (Santiago et al., 2025) |
| moments 1.6.0 | demographic inference (demography) | (Jouganous et al., 2017) |
| demes 0.2.3 | demography interchange | (Gower et al., 2022) |
| ReLERNN 0.2 | recombination map | (Adrion et al., 2020) |
| diploSHIC 1.1.0 | sweep scan | (Kern and Schrider, 2018) |
| RAiSD (`0c705a8`) | sweep corroboration (sweep scan) | (Alachiotis and Pavlidis, 2018) |
| SweepFinder2 1.0 | sweep corroboration (sweep scan) | (Nielsen et al., 2005; DeGiorgio et al., 2016) |
| SLiM 5.2 | forward simulation | (Haller et al., 2026) |
| msprime 1.4.2 | coalescent sim, mutation overlay | (Baumdicker et al., 2022) |
| tskit 1.0.3 | tree-sequence I/O | (Kelleher et al., 2018) |
| pg_gpu 0.1.1.dev14 | windowed $\pi/d_{xy}/F_{ST}$, PCA, local PCA (lostruct method) | (Pope et al., 2026; Li and Ralph, 2019) |
| mkado 0.5.0 | McDonald–Kreitman | (Rivera-Colón et al., 2026) |
| DIAMOND 2.1.12 | homology (inversion, sweep scan) | (Buchfink et al., 2021) |
| AnchorWave 1.2.3 | *I. coindetii*↔*Illex illecebrosus* alignment | (Song et al., 2022) |